Helfi.Space

Manual for users, agents and administrators

Helfi.Space is a lean ticket system for IT departments and institutes. It runs as a single Docker container, signs users in via Active Directory or single sign-on and turns emails into tickets.

Users
Report problems, follow requests, reply – chapter 2

Agents
Work on tickets, templates, reports, keyboard shortcuts – chapter 3

IT service management
Incidents, SLA, problems, changes, knowledge, service catalog, devices, satisfaction – chapter 4

Administrators
Installation, AD, SSO, 2FA, email, updates, backup – chapters 5 and 6

Version 1.12.0 · November 2026
All names, addresses and computers in this manual are examples.

Contents

1. **Overview**
   - Roles
   - Statuses, types and priorities
   - The interface

2. **For everyone: signing in and making requests**
   - Signing in
   - Reporting a problem
   - My requests
   - Replies and files
   - Request participants
   - Emojis
   - Profile, language and password
   - Setting up two-factor authentication
   - Requesting a service
   - Approving
   - Help & knowledge
   - Rating
   - On a smartphone

3. **For agents**
   - Queues and summary
   - Board
   - Ticket list and filters
   - Working on a ticket
   - Replies and internal notes
   - HTML emails
   - Reply templates
   - Several tickets at once
   - Reports
   - Keyboard shortcuts
   - Tickets are never deleted

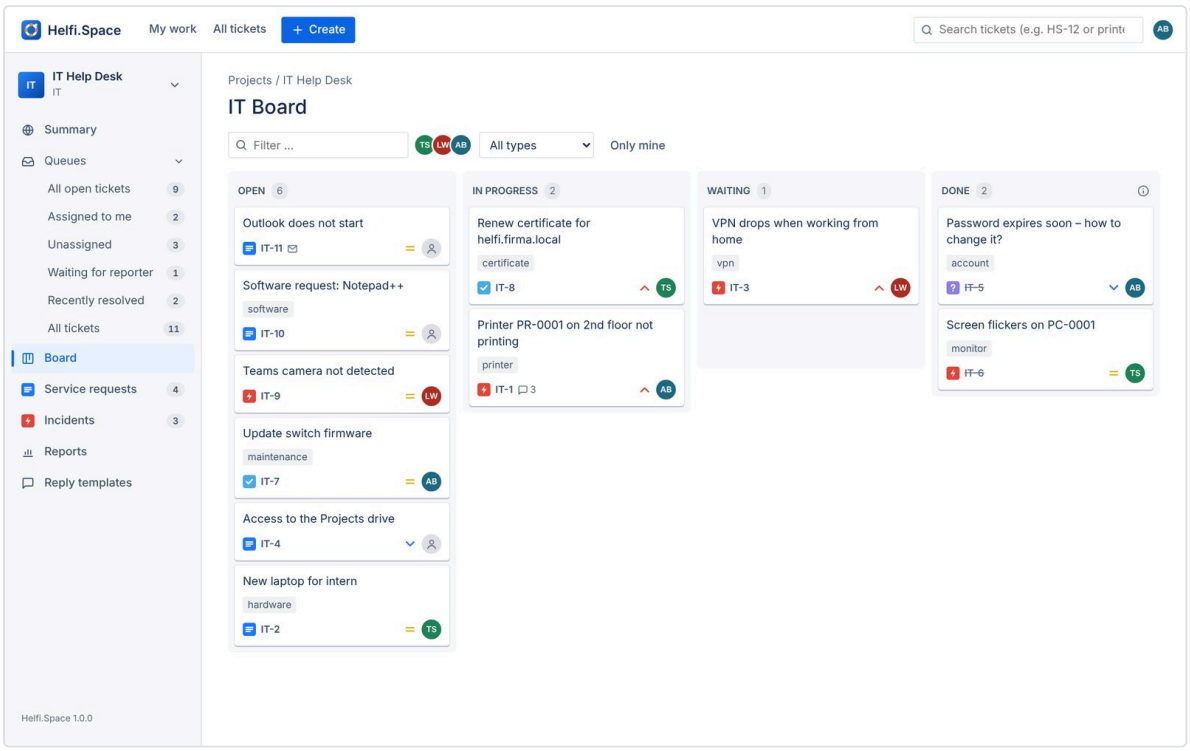
4. **IT service management (ITIL)**
   - Links and subtasks
   - Impact, urgency and major incident
   - Service levels (SLA)
   - Problems and known errors
   - Change management
   - Knowledge base
   - Setting up the service catalog
   - Devices & services
   - Satisfaction survey
   - ITIL key figures

5. **For administrators: setup**
   - Installation
   - HTTPS and reverse proxy
   - Initial setup
   - Active Directory
   - Single sign-on
   - Two-factor authentication and networks
   - Email
   - Projects
   - Users
   - Appearance
   - Sign-in page
   - Import from another ticket system

6. **For administrators: operation**
   - Updates
   - Pinning a version and going back
   - Backup and restore
   - Security
   - Questions and solutions

1 Overview

In Helfi.Space every concern becomes a **ticket** – whether someone reports it in the browser or sends an email to the help desk mailbox. Agents see all tickets of their projects, assign them, reply and finally set them to “Done”. Reporters receive every reply by email and can answer it directly.



Board of a project with the four status columns

1.1 Roles

Role	What it may do
User	Report and follow their own tickets, see tickets they participate in, reply and attach files. Internal notes are never shown to users.
Agent	See and work on all tickets: status, assignee, priority, type, labels, internal notes, reply templates, bulk actions, reports.
Administrator	Everything agents may do, plus *Administration*: projects, users, AD, SSO, security, email, appearance, import, system.

With Active Directory the role is derived from the AD groups at every sign-in (chapter 5.4). The initial setup also creates a **local administrator** that keeps working when AD is unreachable.

1.2 Statuses, types and priorities

Status	Meaning
Open	New or not started yet.
In progress	Someone is working on it.
Waiting	Waiting for the reporter. If they reply, the ticket reopens.
Done	Finished. The reporter gets an email. If they reply to it, the ticket reopens.

Types: Incident (something is broken), Request (something is needed), Task (internal work), Question; for agents also Problem (common cause of several incidents) and Change (planned change), see chapter 4. **Priorities:** Highest, High, Medium, Low.

Every ticket has a **key** made of the project key and a running number, e.g. `IT-42`. It also appears in the subject of all emails, e.g. `[IT-42]`.

1.3 The interface

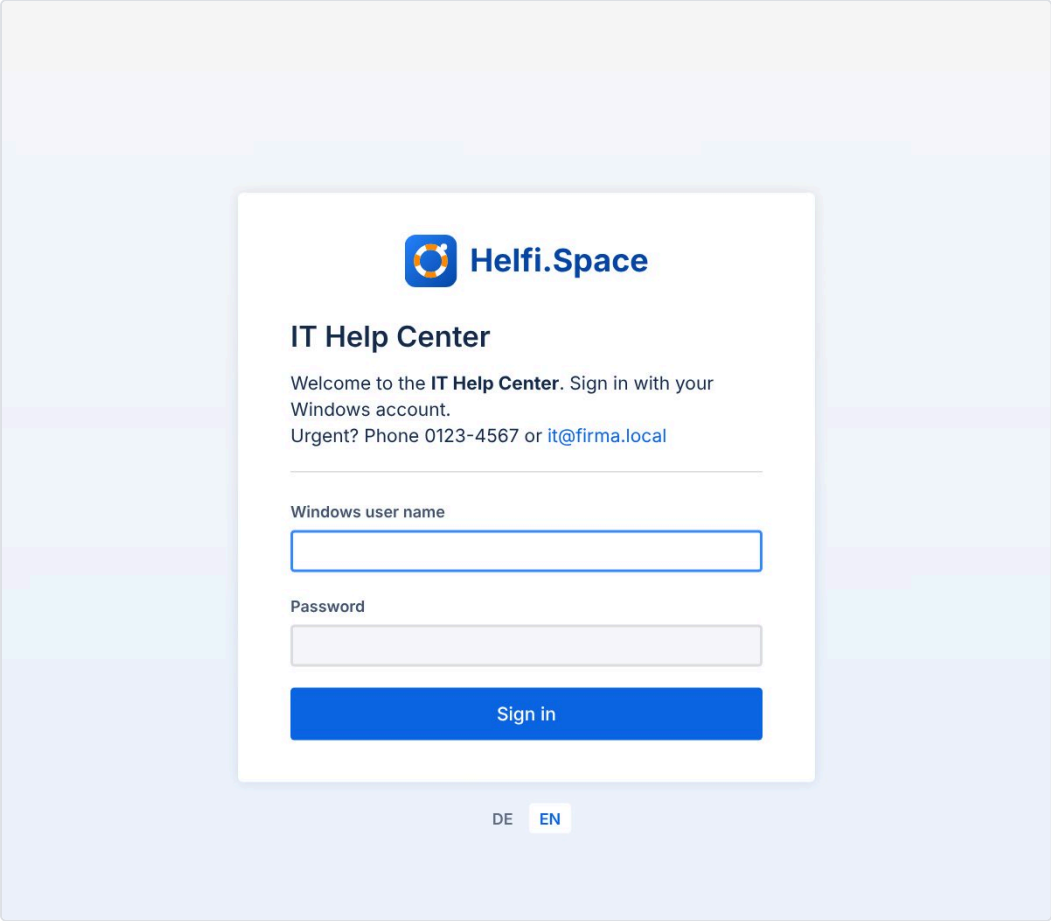
- **Header:** logo, *My work*, *All tickets*, the **Create** button, search (ticket key or keyword) and, on the right, the user menu with profile, language (Deutsch/English) and sign-out.
- **Sidebar (agents):** current project, summary, queues with live counts, board, service requests, incidents, problems, changes, my approvals, change calendar, service catalog, devices & services, knowledge, reports, reply templates and, for admins, settings.
- **Sidebar (users):** *My requests*, depending on the setup *Request a service* and *My approvals*, and *Help & knowledge*.

> The interface is available in German and English. Without a choice of your own it follows the browser language.

2 For everyone: signing in and making requests ALL

2.1 Signing in

Open the address your IT gave you, e.g. `https://helfi.firma.local`. With Active Directory you sign in with your Windows account – as `m.muster`, `FIRMA\m.muster` or `m.muster@firma.local`. If single sign-on is set up, there is an additional button such as “Sign in with company account”, or you are redirected right away.



Sign-in page with a custom welcome text

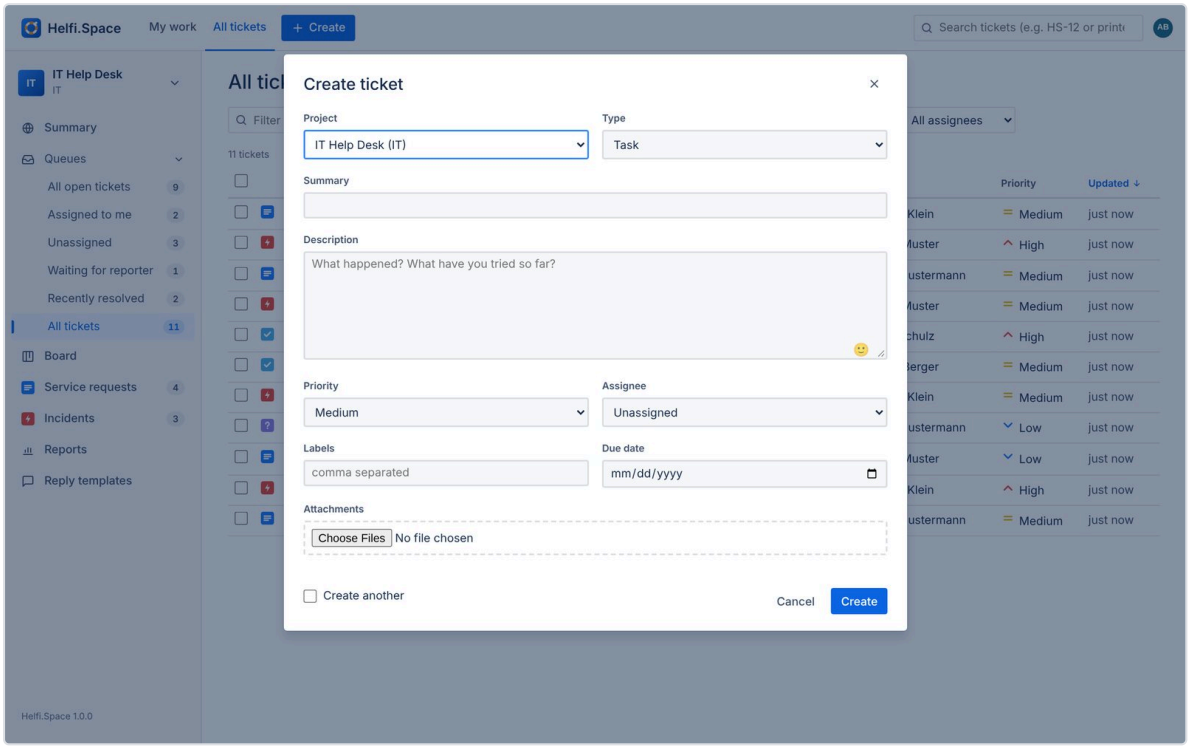
When you sign in **from outside the company network**, Helfi.Space may additionally ask for a 6-digit code from your authenticator app, depending on the settings (see 2.8).

2.2 Reporting a problem

There are two ways:

- **By email** to the help desk mailbox, e.g. `helpdesk@firma.local`. The subject becomes the summary, the text the description, attachments are included. People in *To* or *CC* automatically become participants. You receive a confirmation with the ticket key.

- **In the browser** via **Create** (if your IT allows it): enter summary, description, type and priority, optionally attach files.

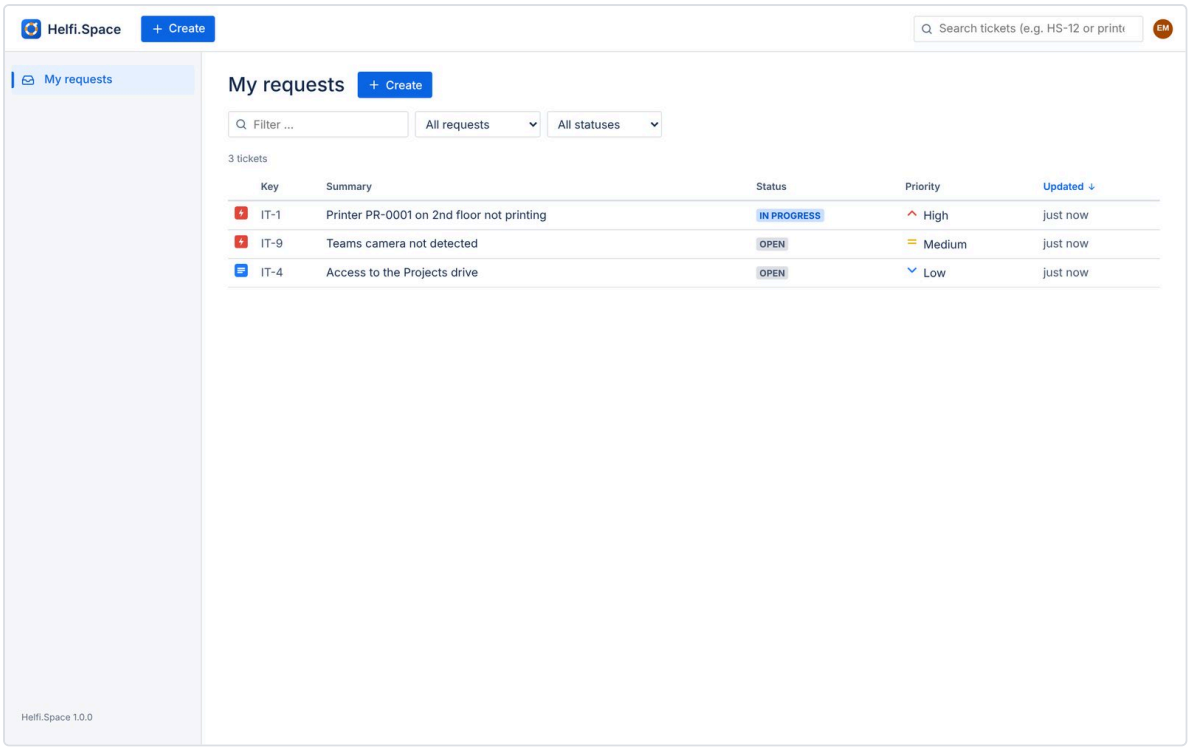


The “Create ticket” dialog

> **Paste screenshots:** an image on the clipboard (e.g. taken with `Win`+`Shift`+`S`) can be pasted with `Ctrl`+`V` directly into the description or a comment.

2.3 My requests

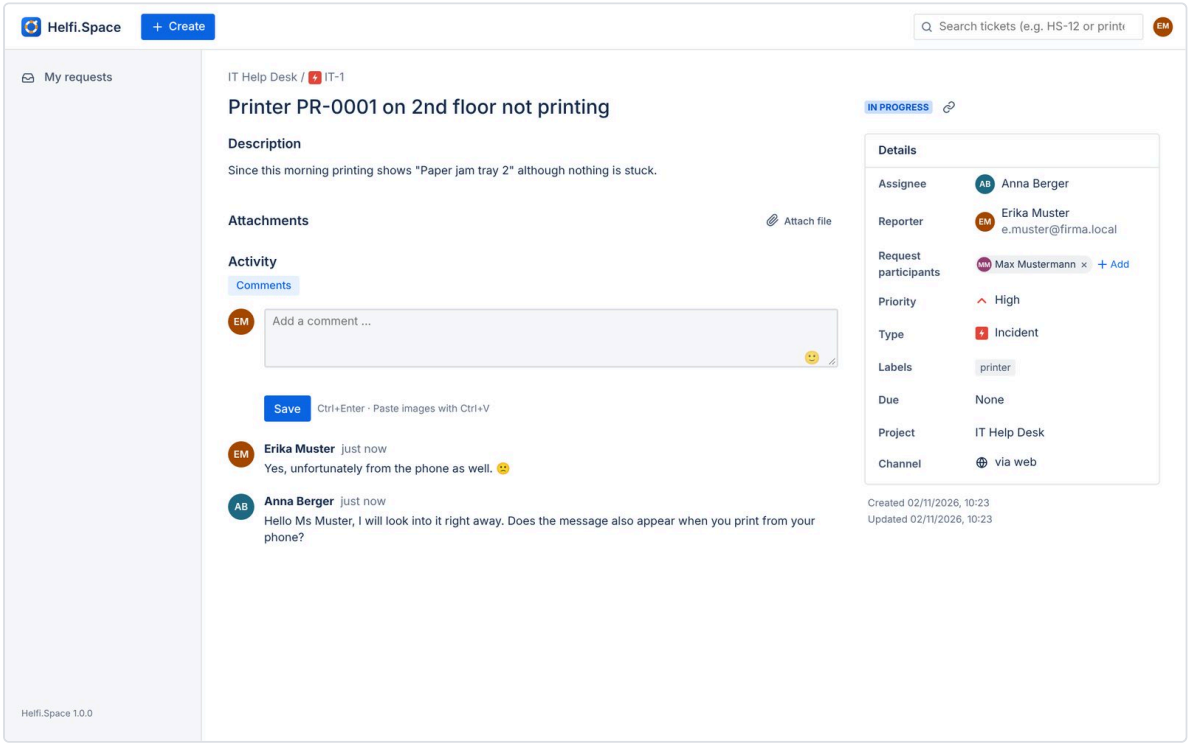
My requests lists all tickets you reported or participate in. Use the filters at the top to narrow them down by status or “I am a participant”.



My requests as seen by a user

2.4 Replies and files

Open a ticket and type into the comment field at the bottom; **Save** or `Ctrl` + `Enter` sends your reply to IT. Alternatively, simply reply to an email from Helfi.Space – the reply lands in the right ticket automatically and quoted history is cut off.



A ticket as seen by the reporter

Attach files under *Attachments* → *Attach file*. Images are shown directly, all other files are downloaded.

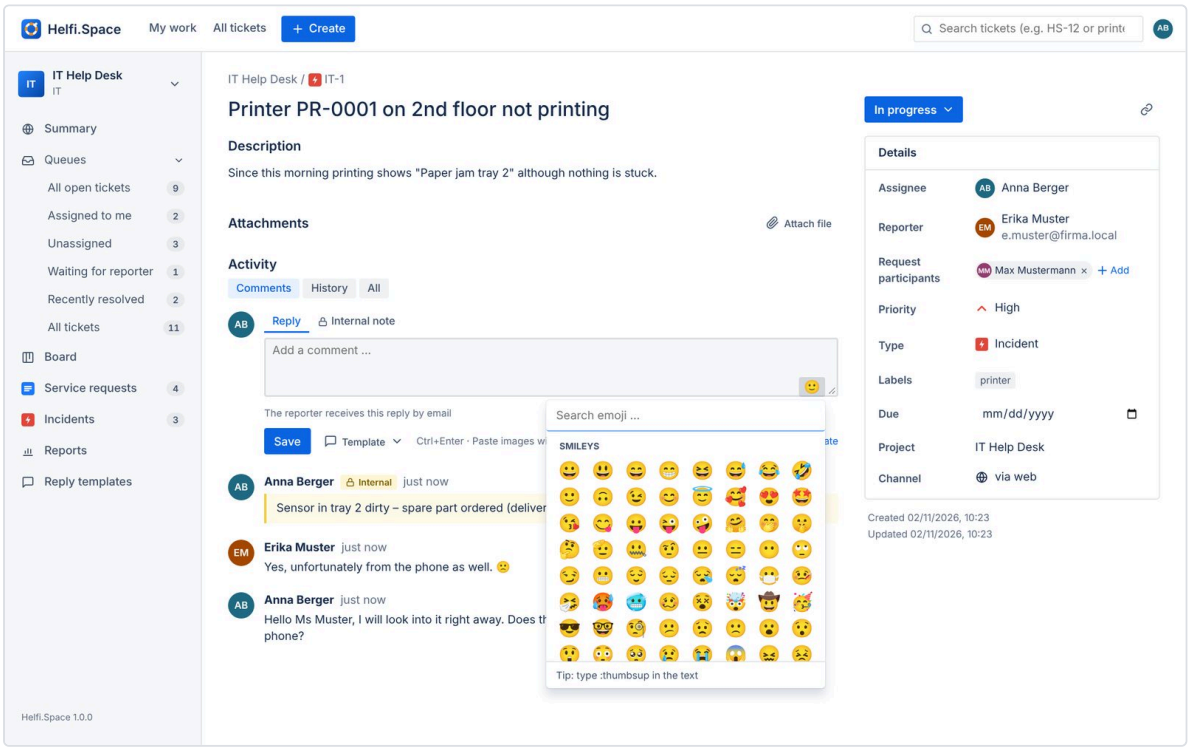
2.5 Request participants

Besides the reporter, further people can take part in a ticket, e.g. colleagues with the same problem. Under *Request participants* → *Add* you add colleagues who have an account (up to 20 per ticket): type a name and the suggestions appear below; pick with a click or `↑`/`↓` and `Enter`, `Esc` cancels.

- Participants see the ticket, can comment and attach files and receive the same emails as the reporter.
- They do not see internal notes and cannot change status, assignee or title. Of the reporter they only see the name, not the email address and logon name.
- Participants can remove themselves; the reporter and agents can remove them too.
- **Agents** can also enter any email address and find people directly in **Active Directory** (marked "AD"), by email address too. The person is taken over with name and email address from AD – they do not need to have signed in before.

2.6 Emojis

- 🙂 **button** at the bottom right of the text field: picker with search (e.g. "thumbs" or "printer"), groups and "Recently used".
- **Short names:** `:thu` suggests 👍 (arrow keys and Enter or Tab to accept, Esc to close). `:tada:` turns into 🎉 right away.
- **Classic smileys** are converted as soon as a space follows: `:)` → 🙂, `;)` → 😉, `:D` → 😄, `:(` → 🙁, `:P` → 😛, `:O` → 😮, `<3` → ❤️, `(y)` → 👍. `Ctrl`+`Z` undoes the conversion right away.

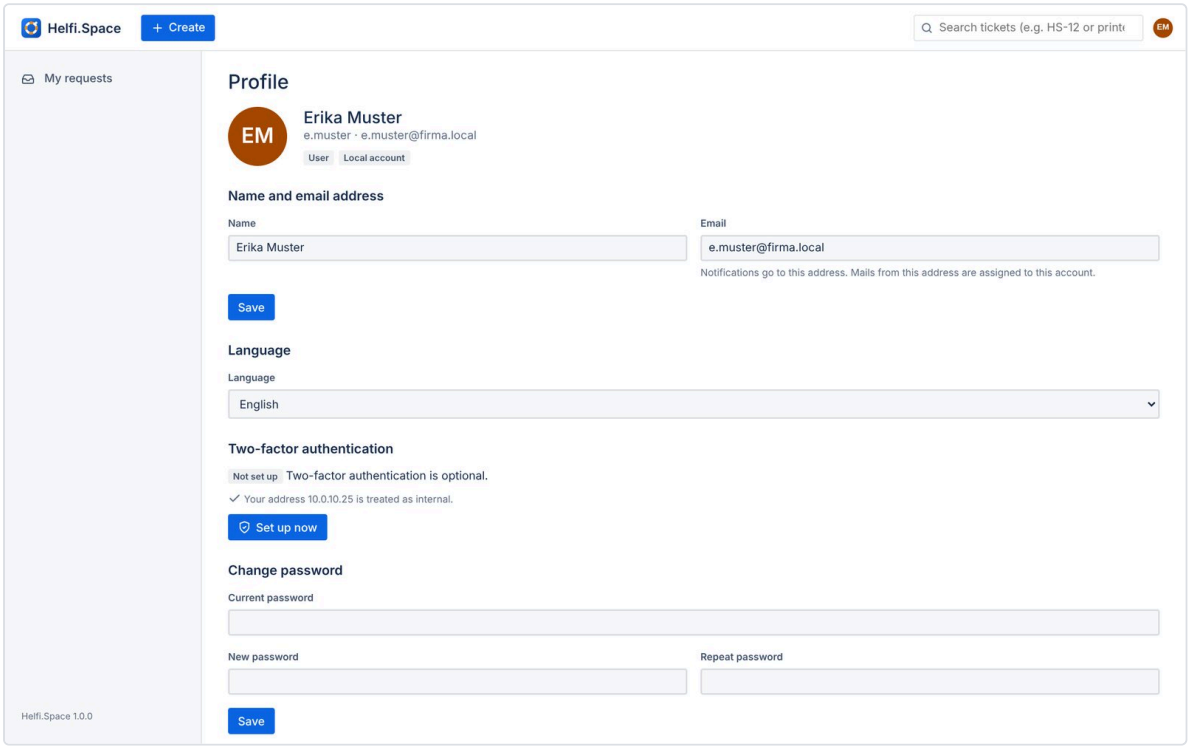


Emoji picker in the comment field

2.7 Profile, language and password

Click your picture at the top right → *Profile* to reach:

- **Name and email address** – editable for local accounts (a new email address requires your current password). For AD and SSO accounts both come from the company account.
- **Language** of the interface and of emails sent to you.
- **What's new** (also in the menu at the top right): what changed in each version. After an update Helfi.Space shows the news once by itself. Clicking the version number at the bottom of the sidebar also leads there, “Helfi.Space” next to it opens the website `helfi.space`.
- **Two-factor authentication** setup and management.
- **Change password** (local accounts only; change AD passwords in Windows as usual).



A user’s profile

2.8 Setting up two-factor authentication

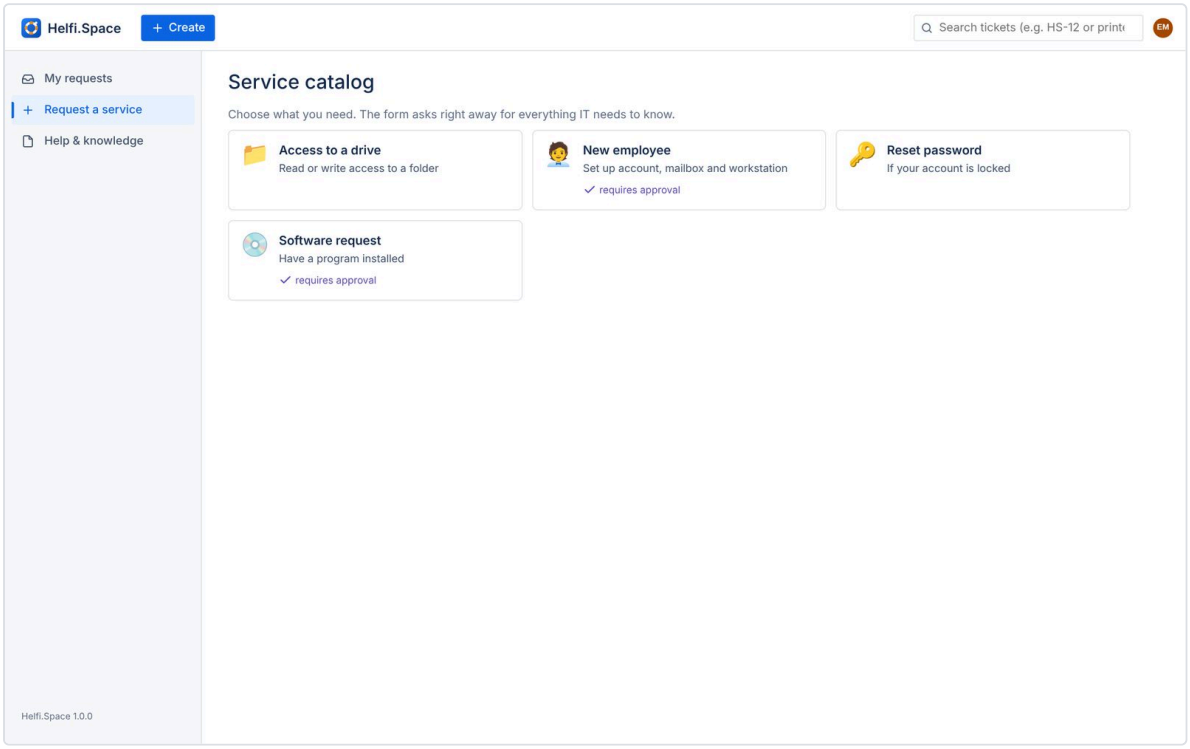
When your IT requires it (or voluntarily): *Profile → Two-factor authentication → Set up now*.

1. Open an authenticator app on your phone (e.g. Microsoft Authenticator, Google Authenticator, Aegis, 2FAS).
2. Scan the QR code.
3. Enter the 6-digit code shown and click **Activate**.
4. Keep the **10 recovery codes** safe (print them or store them in a password manager). Each works exactly once if your phone is not at hand.

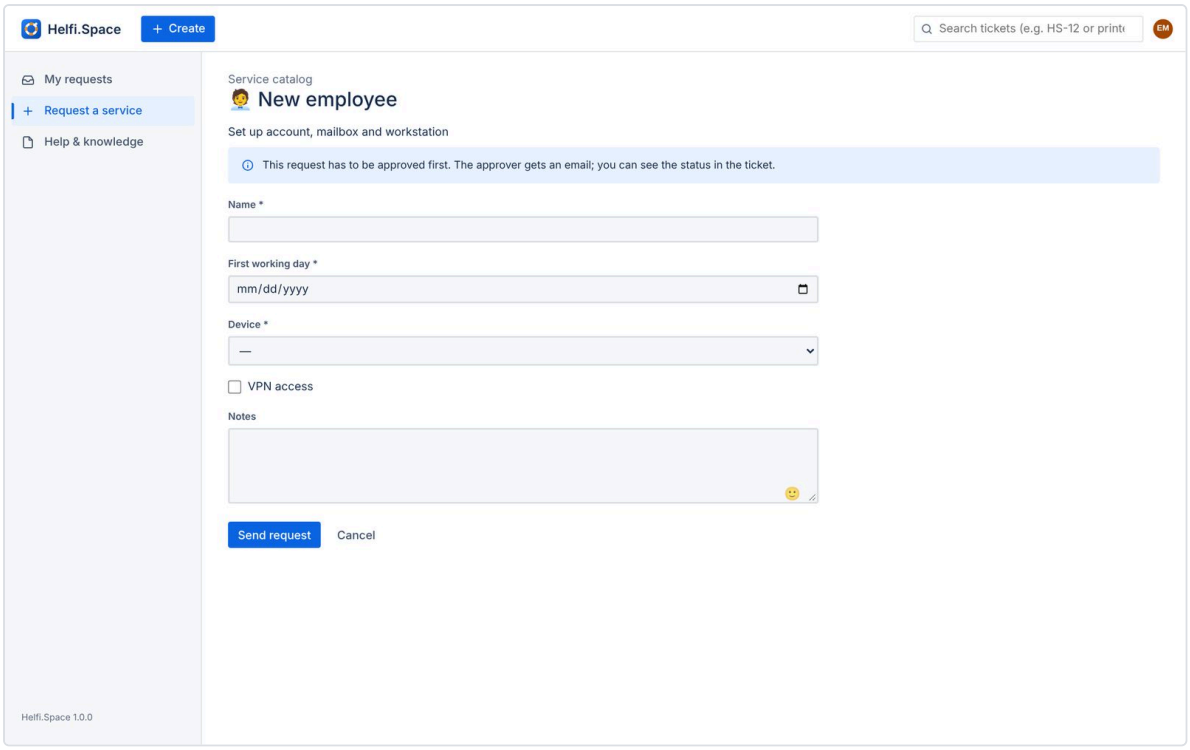
> For security reasons, first-time setup normally only works **inside the company network** (except after an SSO sign-in whose second factor was checked by the identity provider). This way nobody with a stolen password can register their own second factor from outside. **Lost your phone?** Sign in with a recovery code or ask IT to reset your 2FA.

2.9 Requesting a service

For frequent requests there are ready-made forms, e.g. “New employee” or “Software request”: *Request a service* in the sidebar. Clicking a tile opens the form; required fields are marked with *. After sending you land in the new ticket.



Service catalog as a user sees it

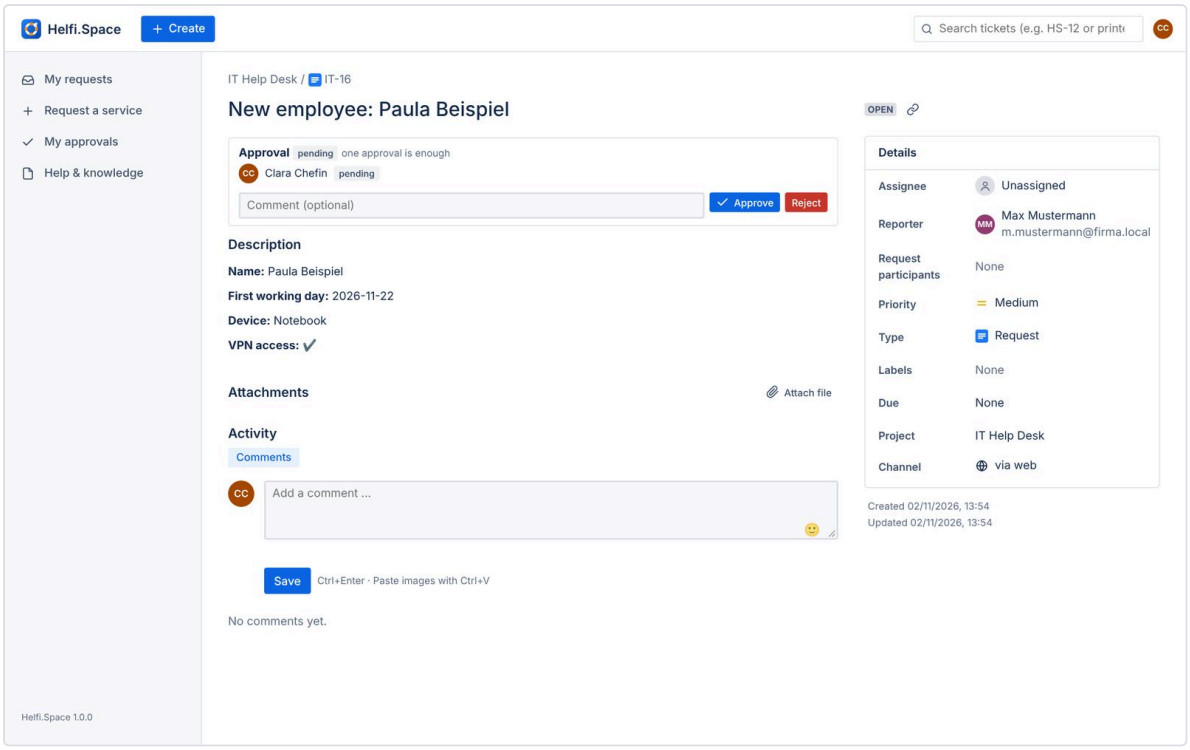


Form “New employee” – this request needs approval

If an offer says “requires approval”, someone has to agree first, e.g. your manager. You can see the status at the top of the ticket.

2.10 Approving

Approvers get an email and see *My approvals* in the sidebar. In the ticket one click on **Approve** or **Reject** is enough, optionally with a comment. The decision is sent to the reporter as a reply; a rejection closes the request. Approvers without the agent role only see the requests they decide on, and only the reporter’s name.

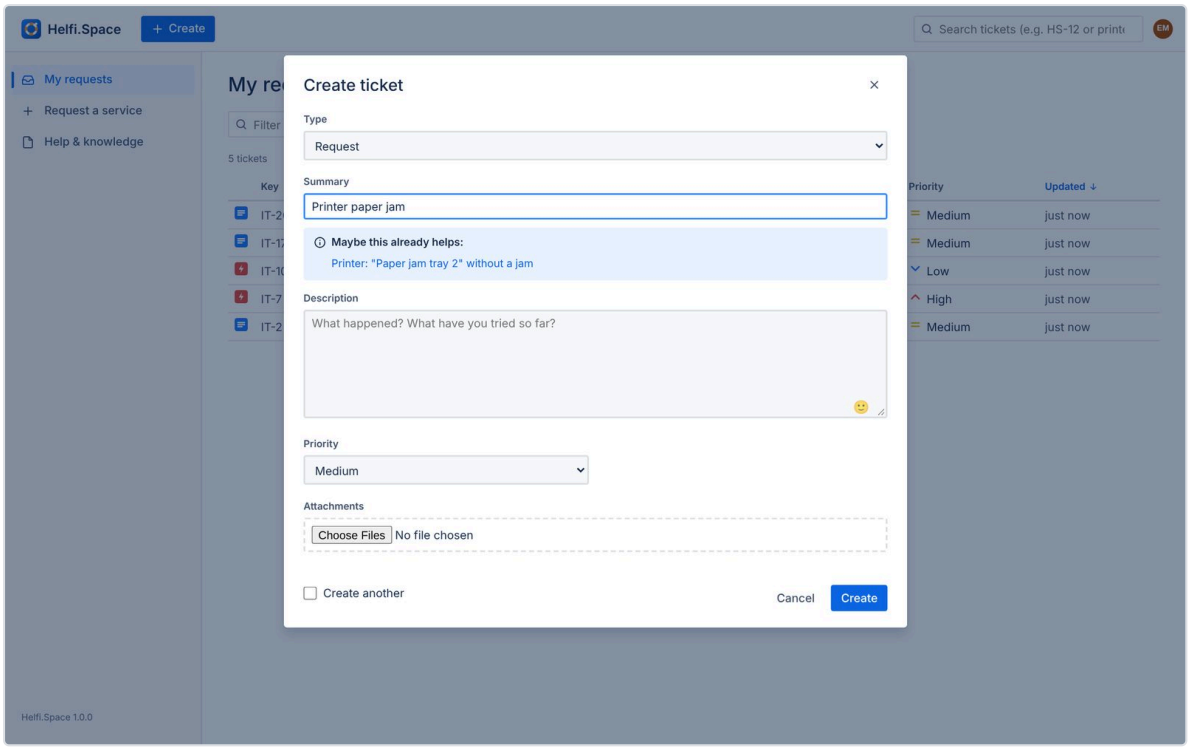


Approval in the ticket as the manager sees it

2.11 Help & knowledge

Under *Help & knowledge* you find guides from IT, e.g. “Set up VPN when working from home”. While you report a problem, Helfi.Space already suggests matching articles (“Maybe this helps already”) as soon as you type the summary. A click opens the article without losing your report.

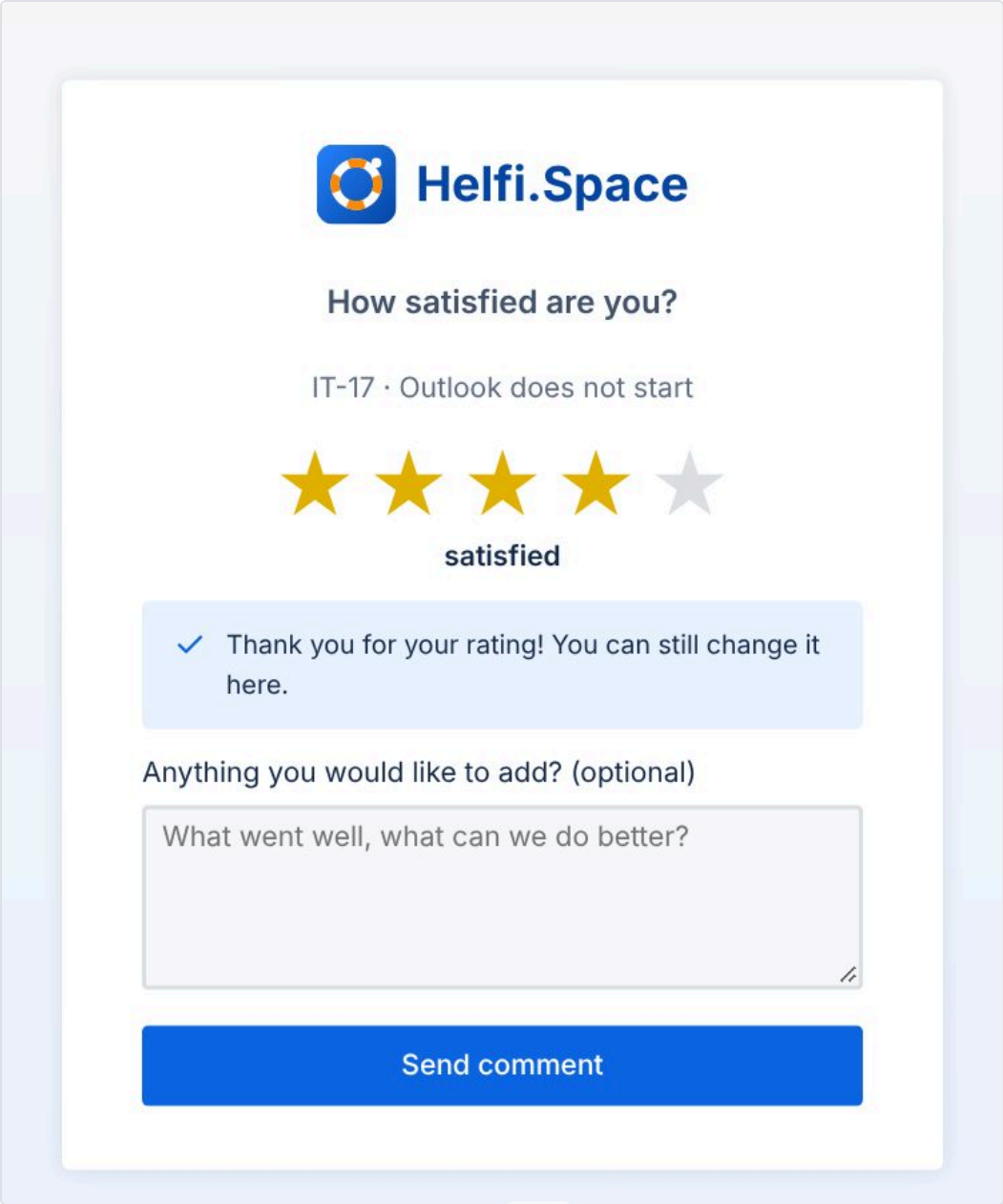
Below each article you can write a **comment** if something is missing or out of date – IT gets notified. **Print / PDF** prints the article or saves it as a PDF.



Knowledge suggestion while reporting

2.12 **Rating**

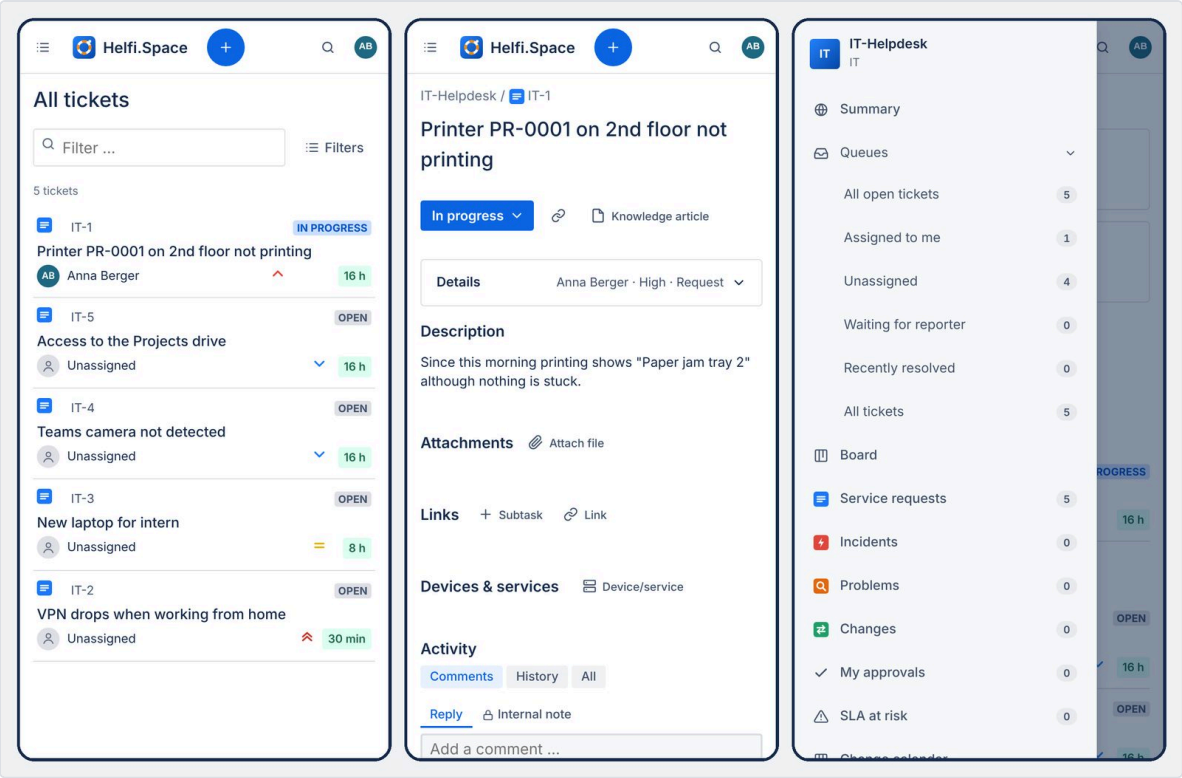
When your ticket is done, IT may ask you for a short rating. The “Resolved” email contains five links from “very satisfied” to “very dissatisfied”: one click is enough, no sign-in needed. On the page you can still change the stars and optionally write a comment. When signed in, you can also rate directly in the ticket.



Rating page after one click in the email

2.13 On a smartphone

Helfi.Space adapts to narrow screens. Everything that works on the PC also works on the phone.



Ticket list, ticket and menu on a smartphone

- **Menu:** ☰ at the top left opens the sidebar with all areas. The **magnifier** opens search, the round ＋ creates a ticket.
- **Lists** show each ticket as a card. The selection is behind **Filters**. Agents select tickets for bulk actions by **pressing and holding a ticket**; after that each tap selects another one.
- **Ticket:** title and status come first, then the **details** as an expandable line with assignee, priority and type, then description and activity.
- **Board:** swipe sideways through the columns. Change the status with ⋯ on the card.
- **Administration:** choose the section from a list at the top.

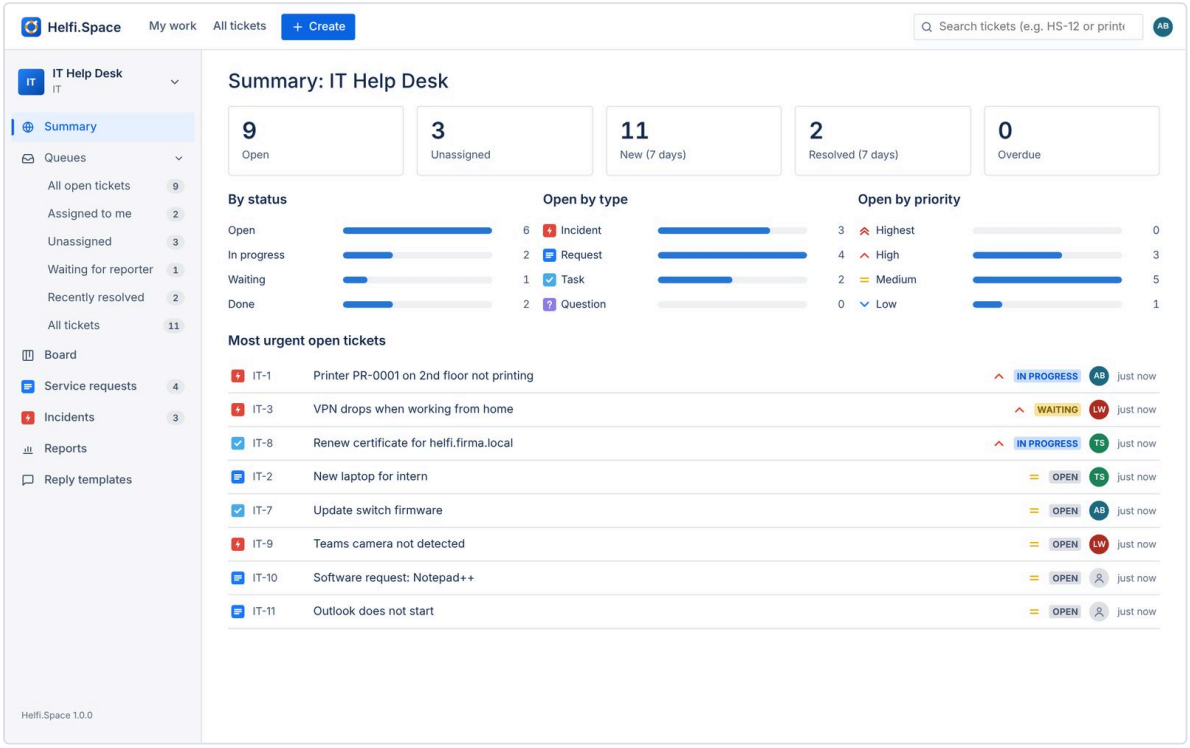
> **Like an app:** choose *Add to Home Screen* in the browser menu (iPhone: share icon, Android: ⋮). Helfi.Space then starts full screen with its own icon. It still needs an internet connection.

3 For agents AGENTS

3.1 Queues and summary

For the selected project, the sidebar shows fixed **queues** with live counts: *All open tickets*, *Assigned to me*, *Unassigned*, *Waiting for reporter*, *Recently resolved* (7 days) and *All tickets*. Below them *Service requests* and *Incidents* (open requests and incidents). Switch the project at the top of the sidebar.

The **summary** shows key figures, distributions by status, type and priority and the most urgent open tickets.



Project summary

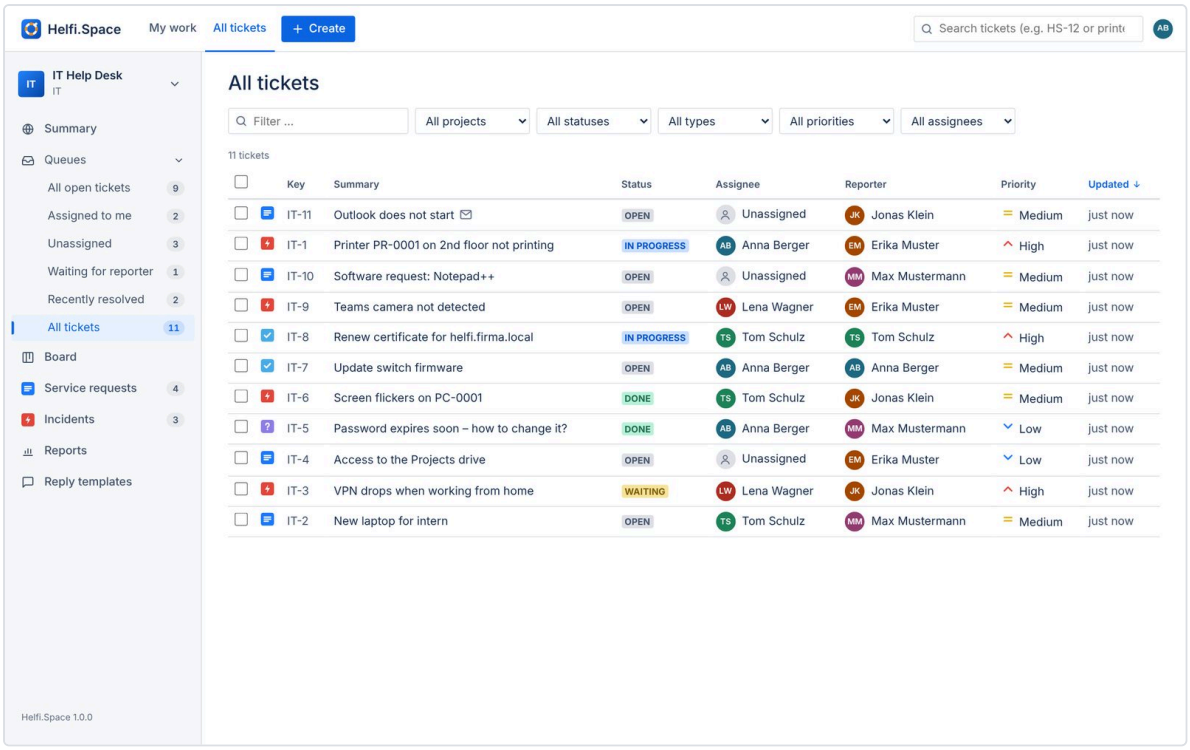
My work in the header collects everything assigned to you, everything unassigned and everything you reported yourself.

3.2 Board

The board shows the tickets in four columns by status. Drag a card into another column to change its status, or within a column to set the order. At the top you can filter by text, assignee (avatars), type or “Only mine”.

3.3 Ticket list and filters

All tickets shows a sortable list. Filters for project, status, type, priority and assignee can be combined; clicking a column header sorts. The page address contains the filters – bookmark or share it.



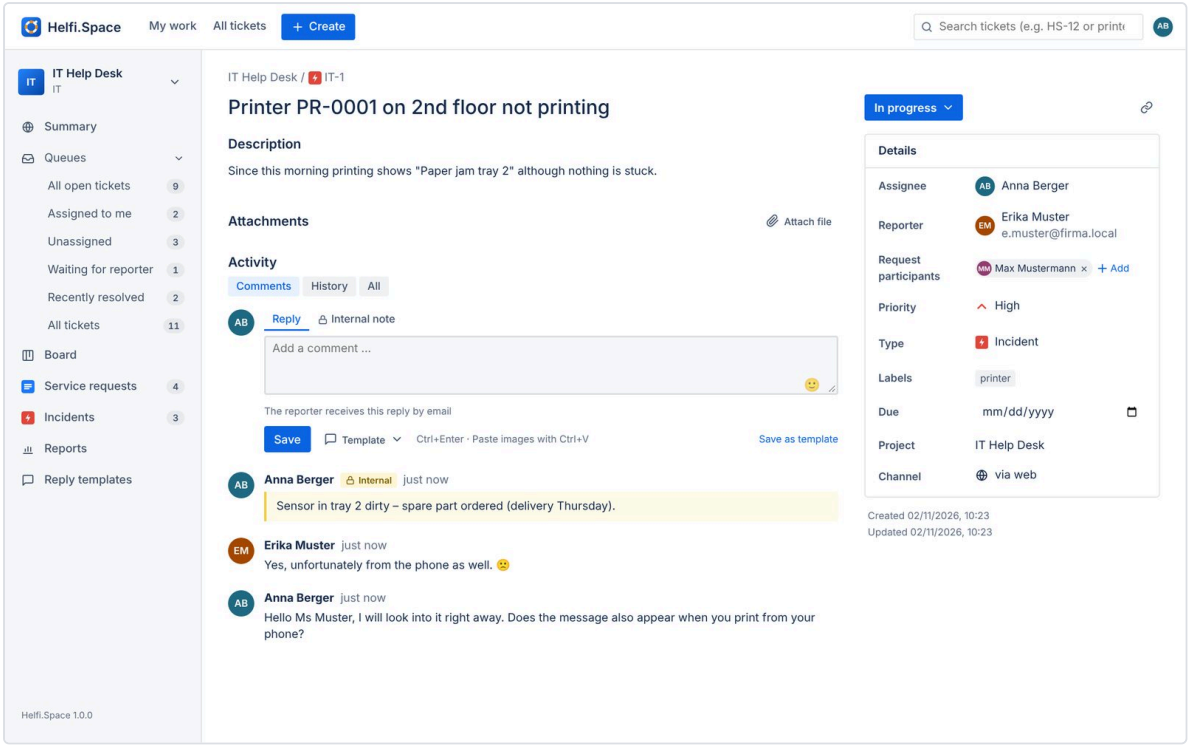
Ticket list with filters

Search

The search box at the top searches the full text of all tickets: title, description, replies, labels and attachment names. Agents also find internal notes and the people involved (also by email address and logon name). Users without the agent role find tickets by title, description, public replies, attachment names, labels and the names visible in the ticket – not by email addresses or logon names.

- **Word parts** match: “printer” also finds “networkprinter”.
- **Several words** must all appear, in any order. `"exact phrase"` in quotes, `-word` excludes.
- Results are sorted by **relevance**, title matches first. Each result shows where it matched with the words highlighted.
- A ticket number jumps straight to the ticket.

3.4 Working on a ticket



Ticket view: description and activity on the left, status and details on the right

- **Status** via the button at the top right (e.g. “Open” → “In progress”).
- Change **assignee**, **priority**, **type**, **labels**, **due date** and **project** directly in the details. “Assign to me” takes one click or the `i` key.
- Click **title** and **description** to edit them.
- **Changing the project:** the ticket gets a new key in the new project; the old one stays in the history. Replies to old emails are still assigned correctly.
- **Activity:** tabs *Comments*, *History* (every change with person and time) and *All*. For a changed description, **Show change** shows the differences and the previous version.
- **Copy link** via the chain icon at the top right.

3.5 Replies and internal notes

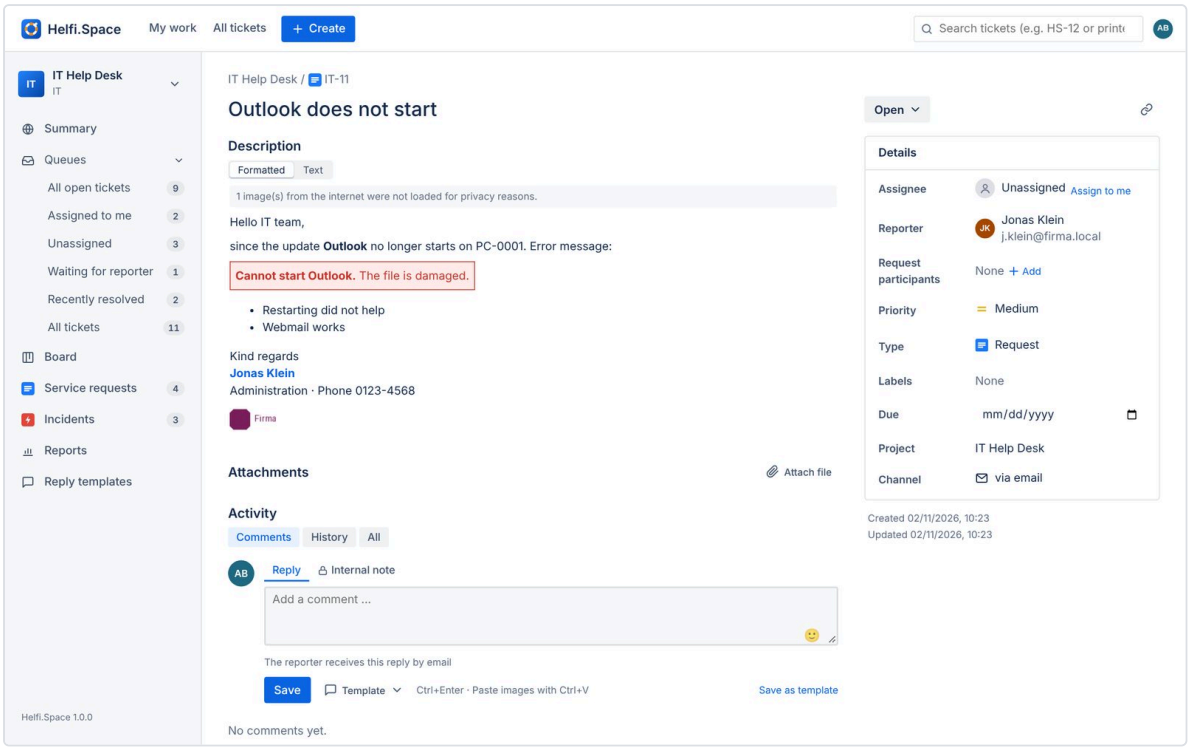
Above the comment field choose between **Reply** and **Internal note**:

- **Replies** are emailed to the reporter and all participants, each in their own language.
- **Internal notes** (yellow background) are only visible to agents. They are never sent by email, and pasted images stay internal as well.

> Emails coming from an agent’s address never count as agent replies (sender addresses can be forged). Agents therefore always reply via the web interface. Email replies to assignment, SLA or approval mails sent to agents become internal notes and do not reopen the ticket.

3.6 HTML emails

When a ticket or a reply arrives as an HTML email, Helfi.Space shows it **formatted** – with tables, bold text, colours, signature and embedded images. The **Formatted | Text** switch changes to the text version; the browser remembers the choice.

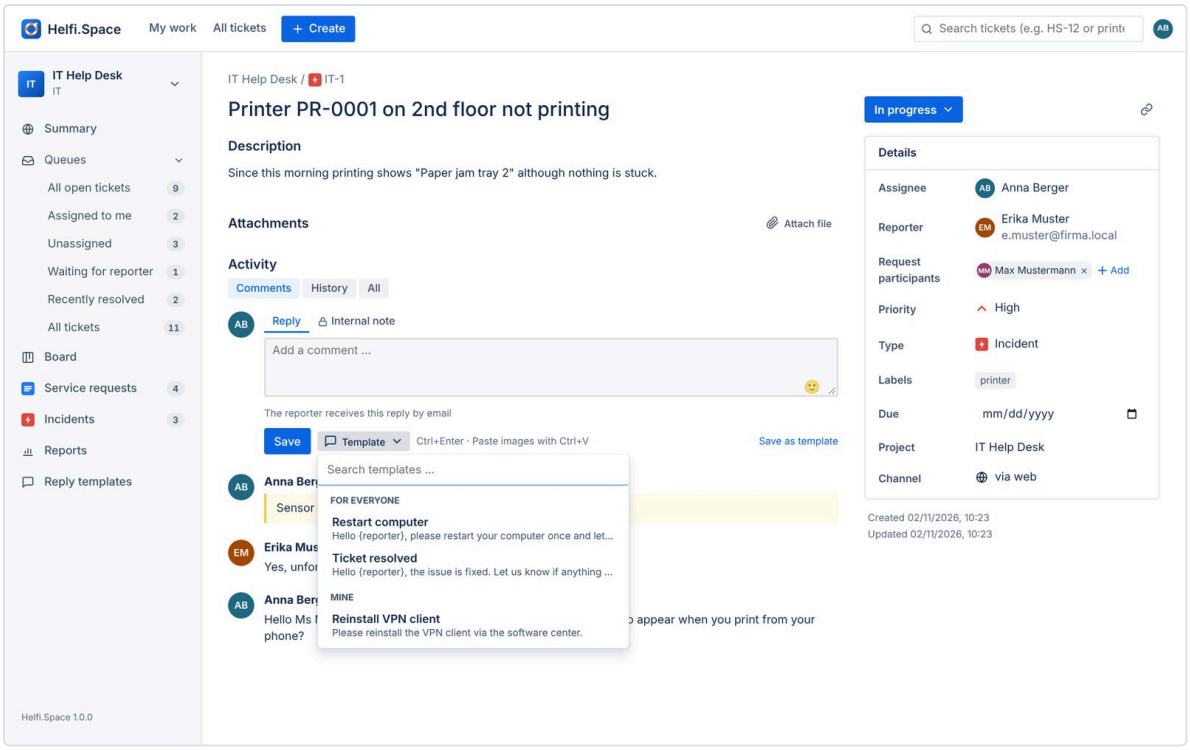


Formatted view of an HTML email

- Images from the internet (often tracking pixels) are never loaded for privacy reasons; a notice states how many there were.
- In replies the quoted history is collapsed (“Show quoted text”).
- Editing always uses the text version. After a description has been edited, only the text applies.

3.7 Reply templates

For replies that are always the same: click **Template** in the comment field, type to search, insert with Enter or a click. The text lands at the cursor position and can still be adjusted before saving.



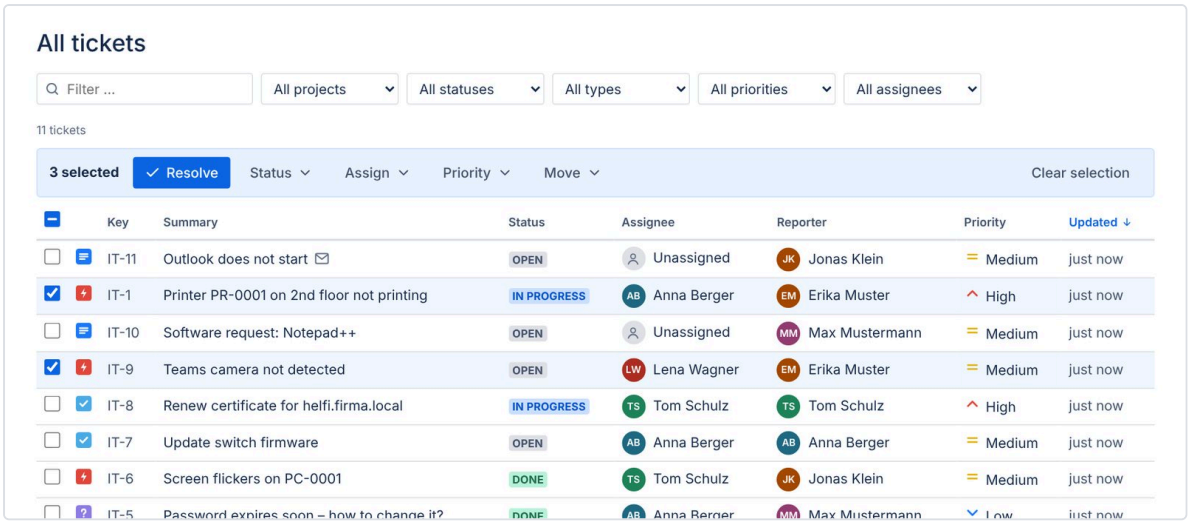
Choosing a template: “For everyone” at the top, your own below

- **Placeholders** are replaced on insertion: `{reporter}`, `{ticket}`, `{title}`, `{agent}`.
- **New template:** write the text in the comment field and choose *Save as template*. Personal templates are managed in your *Profile*; templates for everyone are maintained by admins under *Administration → Reply templates*.

3.8 Several tickets at once

In *All tickets* select tickets with the check boxes (Shift+click for a range, `x` for the highlighted ticket). The action bar offers:

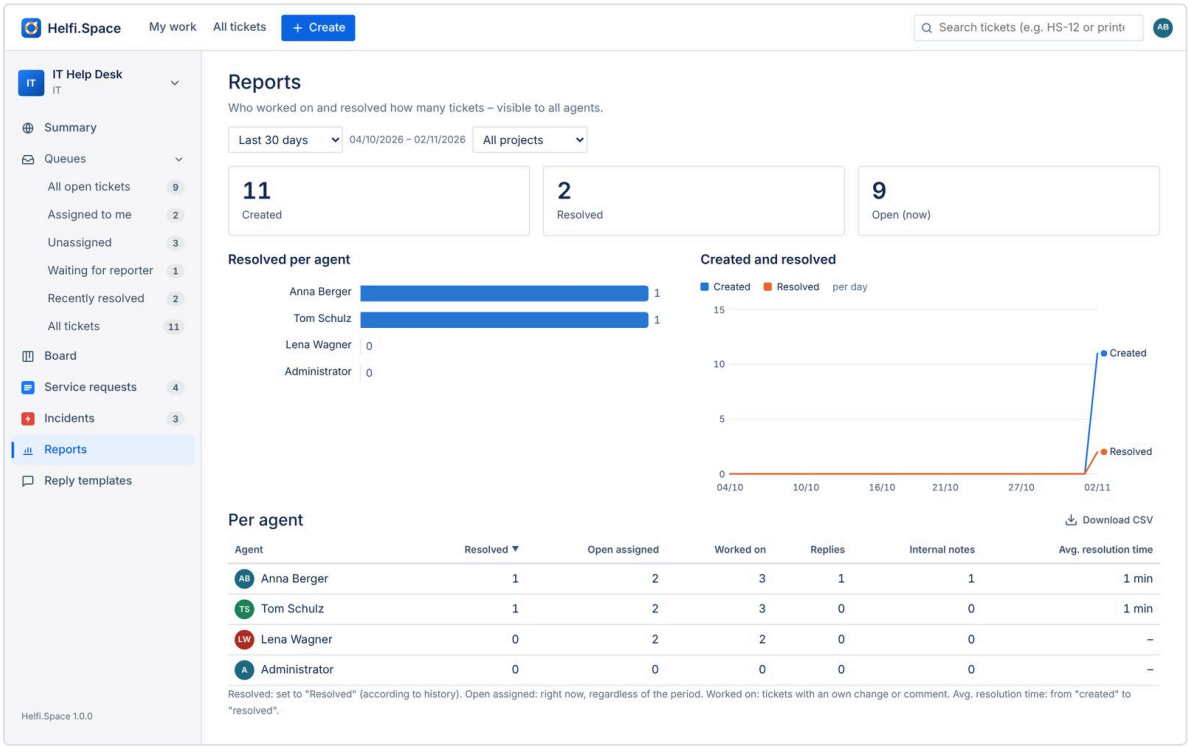
- **Resolve** – optionally with a reply to reporter and participants, also with a template and placeholders per ticket.
- **Status**, **Assign**, **Priority** and **Move** to another project.



Three tickets selected – action bar at the top

3.9 Reports

Reports shows, for a period (7, 30, 90 days, this year or custom) and optionally one project: created, resolved and open tickets, “Resolved per agent”, the “Created and resolved” trend and a table per agent (resolved, open assigned, worked on, replies, internal notes, average resolution time). The table can be sorted and downloaded as CSV.



Reports

3.10 Keyboard shortcuts

? shows all shortcuts. They never fire while you are typing in a field.

Where	Key	Action
everywhere	`c`	create ticket
	`/`	search
	`g` then `b` / `i` / `m`	board / all tickets / my work
ticket	`i`	assign to me
	`a`	choose assignee
	`m`	comment
	`e` / `l`	edit title / labels
	`.`	actions with search (status, priority, type, template …)
	`Ctrl`+`Enter`	save comment
lists	`j` / `k`	next / previous ticket
	`o` / `Enter`	open highlighted ticket
	`x`	select highlighted ticket

`Enter` on a focused link or button activates it. `Enter` only opens the row highlighted with `j`/`k` when the focus is not on a control.

3.11 Tickets are never deleted

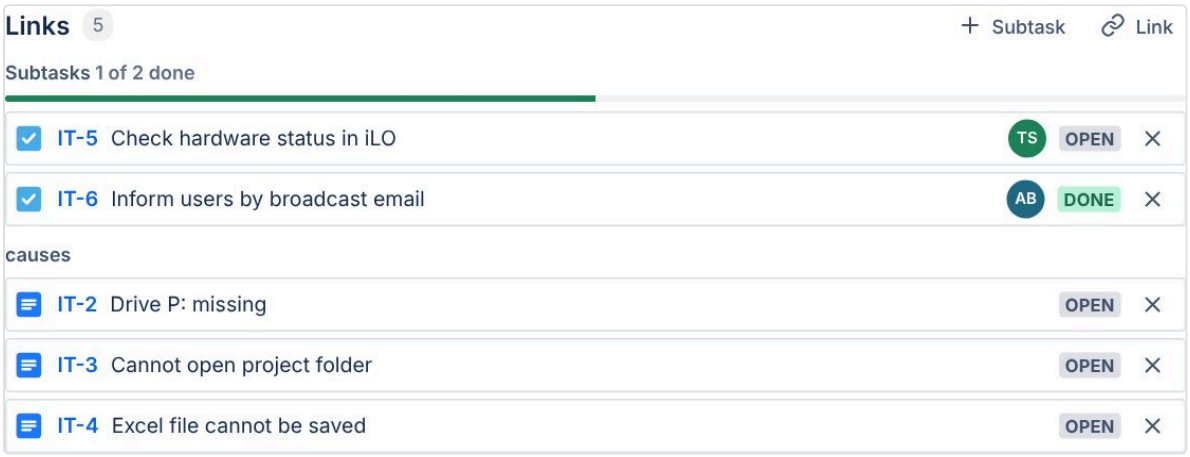
Tickets are records: who reported what and when, and what was done? That is why they **cannot be deleted**, not even by administrators. Finished work is set to “Done”. Against spam, use a sender rule “Ignore” (5.7) – then no ticket is created in the first place.

4 IT service management (ITIL) AGENTS

Helfi.Space covers the most important ITIL processes without forcing you to use all of them: incidents and major incidents, service levels, problems, changes, knowledge, service catalog, devices and services, and satisfaction. Whatever is not switched on or set up stays invisible.

4.1 Links and subtasks

In the *Links* section of a ticket you connect tickets: “relates to”, “duplicates” (optionally resolving the duplicate right away), “blocks”, “causes” and “is subtask of”. **Subtask** directly creates a task with an assignee; a bar shows how many are done.

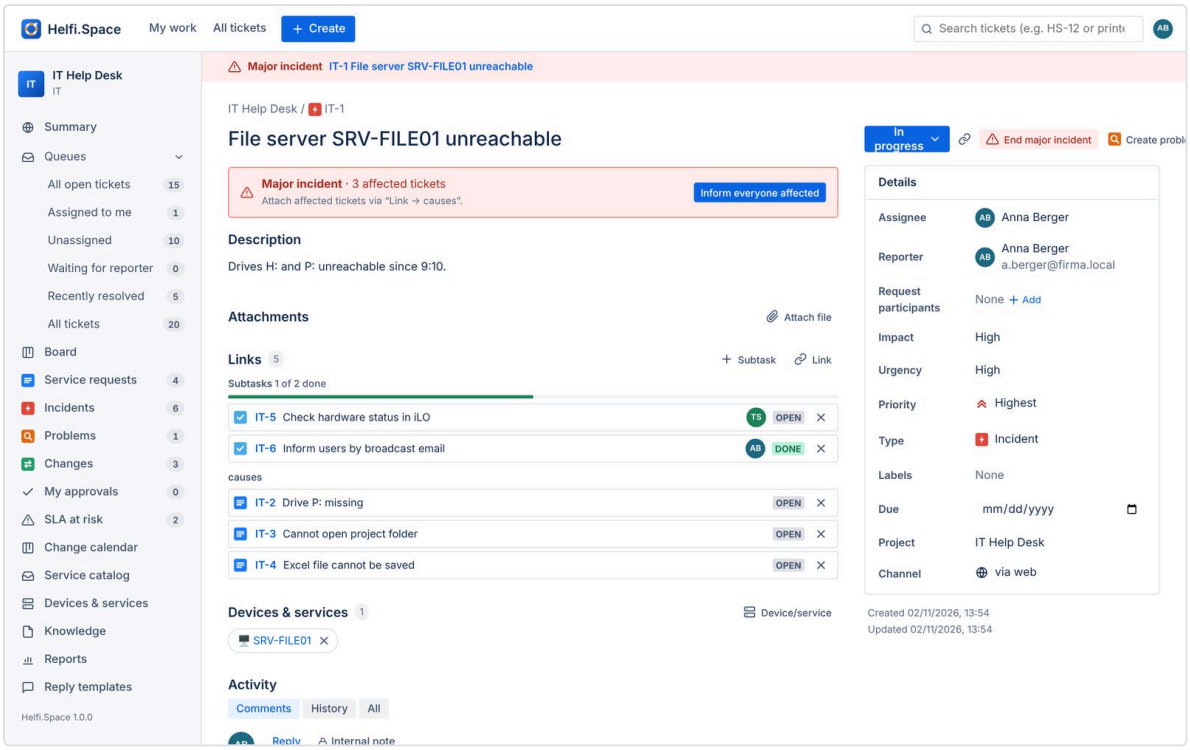


Subtasks with progress and caused incidents

4.2 Impact, urgency and major incident

In the details you set **impact** (how many are affected?) and **urgency** (how fast must it be fixed?). When both are set, the priority follows the ITIL matrix, e.g. high/high = Highest.

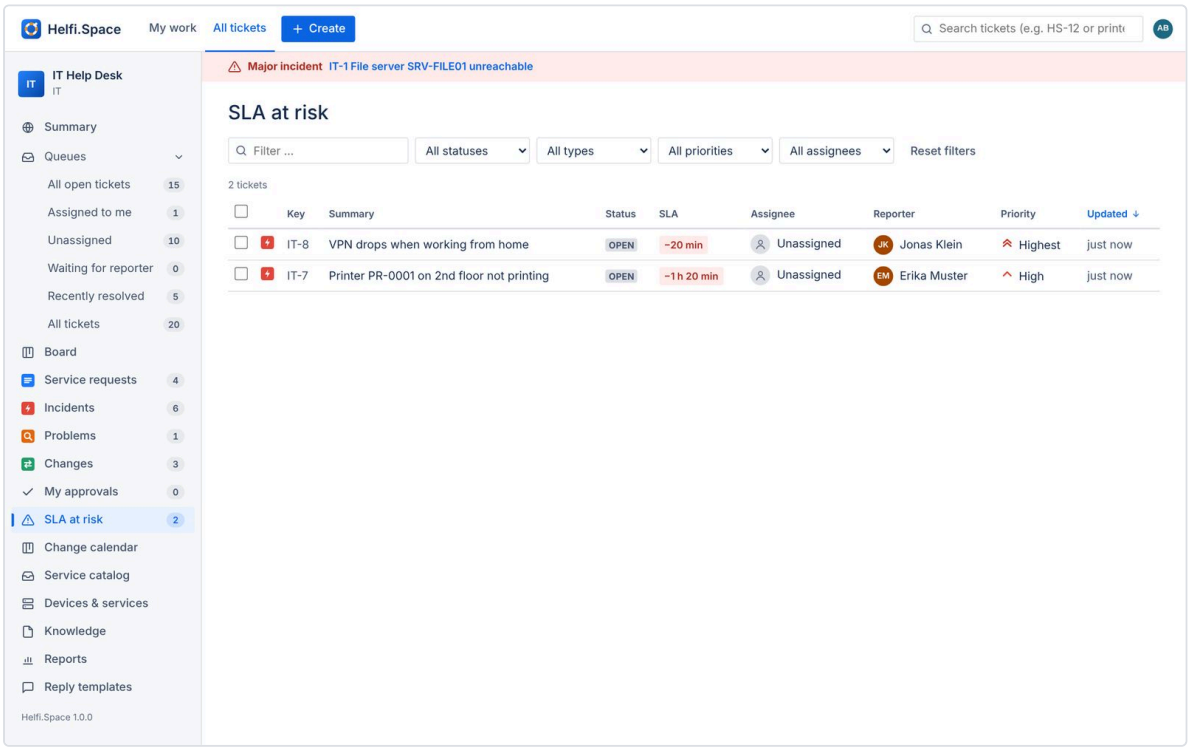
Mark a severe incident affecting many people as a **major incident**. All agents then see a red bar at the top. Link individual reports with “causes”. **Inform everyone affected** sends a message to the reporters and participants of all open linked tickets; with “resolve” they are resolved together.



Major incident with subtasks, affected tickets and device

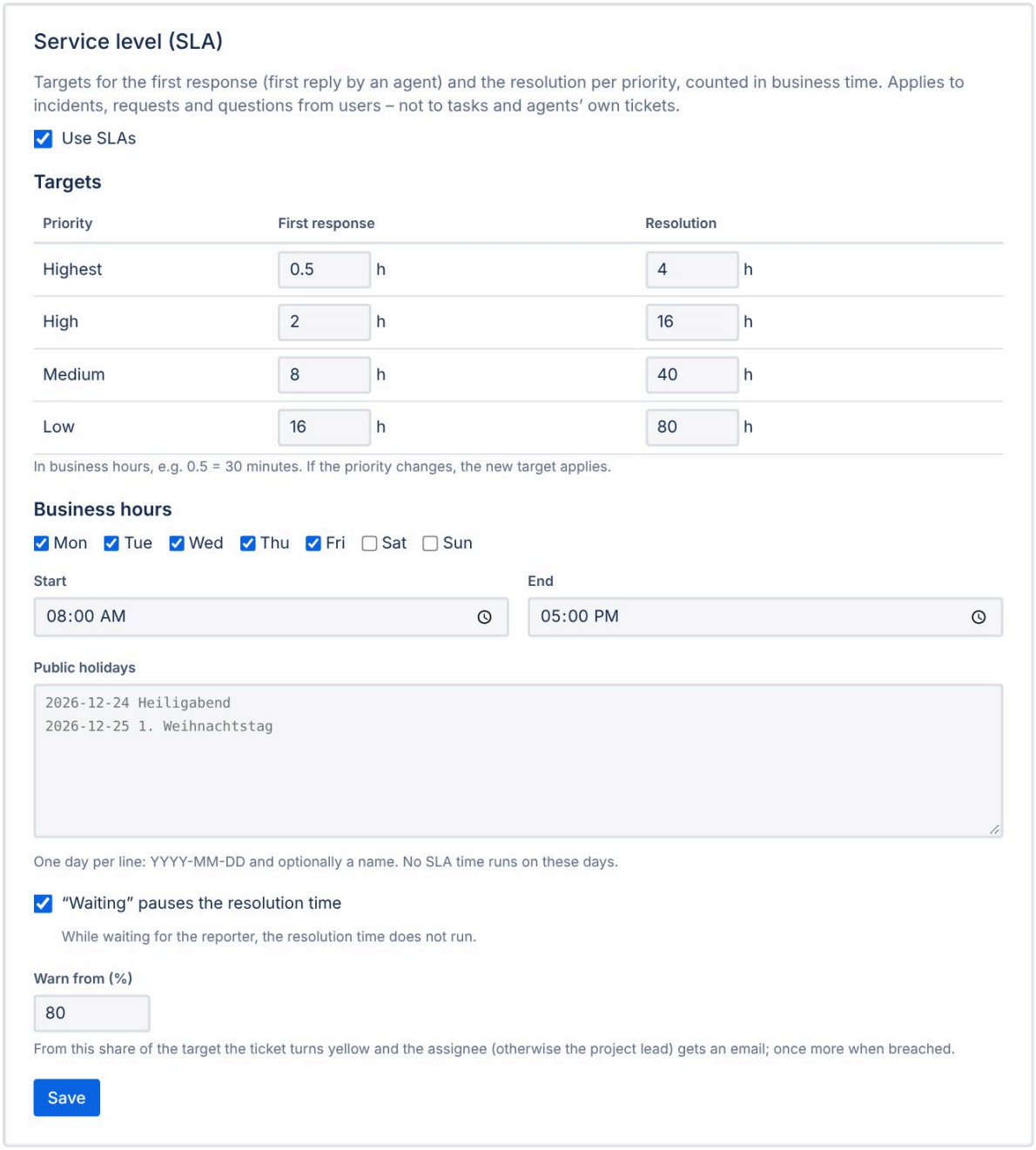
4.3 Service levels (SLA)

With SLAs switched on, every ticket from users has two clocks: **first response** (until IT's first public reply) and **resolution**. Both only run during business hours; holidays and, optionally, the status "Waiting" pause them. At 80 % of the time the ticket turns yellow and the assignee (otherwise the project lead) gets an email; when the target is exceeded it turns red. The queue *SLA at risk* shows all tickets running short.



Queue "SLA at risk" with remaining time

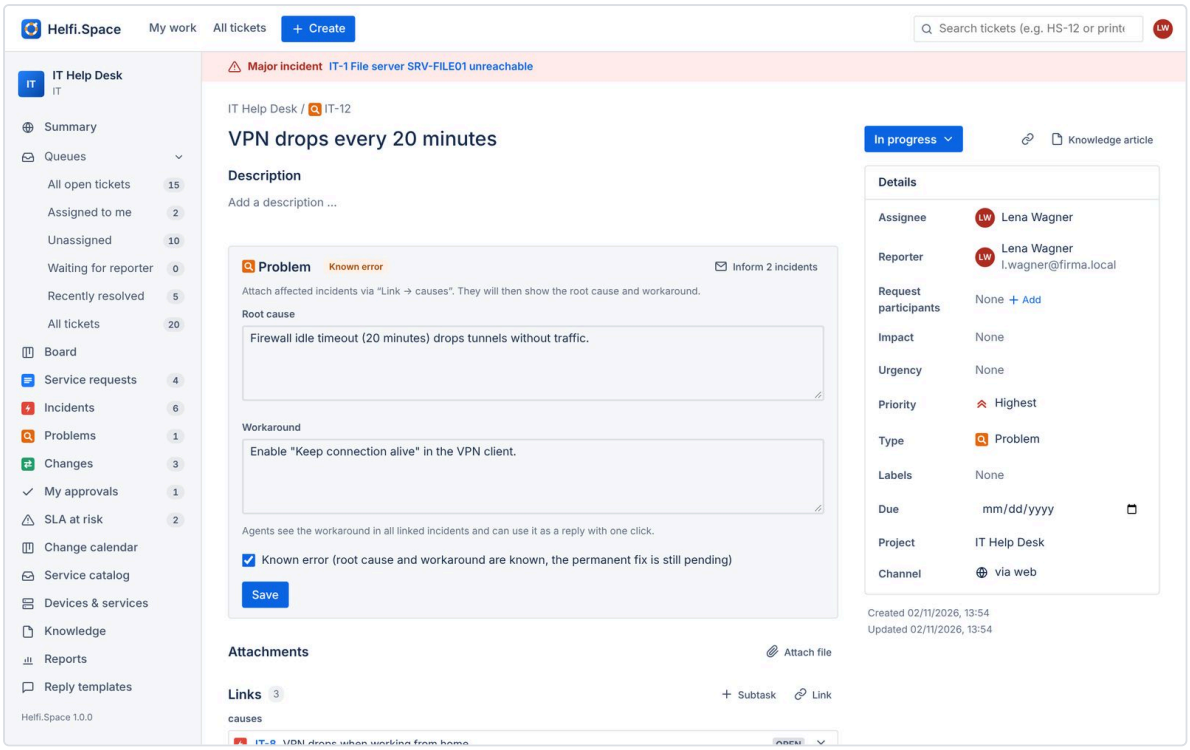
ADMIN Targets per priority, business hours, holidays and warning threshold under *Administration → Service level*. Default: off.



Administration → Service level

4.4 Problems and known errors

When the same incidents pile up, there is usually a common cause. **Create problem** in the ticket creates a problem and links the incident. Attach further incidents with "causes". In the problem you record the **root cause** and **workaround**. Marked as a **known error**, agents see the workaround in all linked incidents and can send it as a reply with one click. The tab *Known errors* under *Knowledge* lists them all.



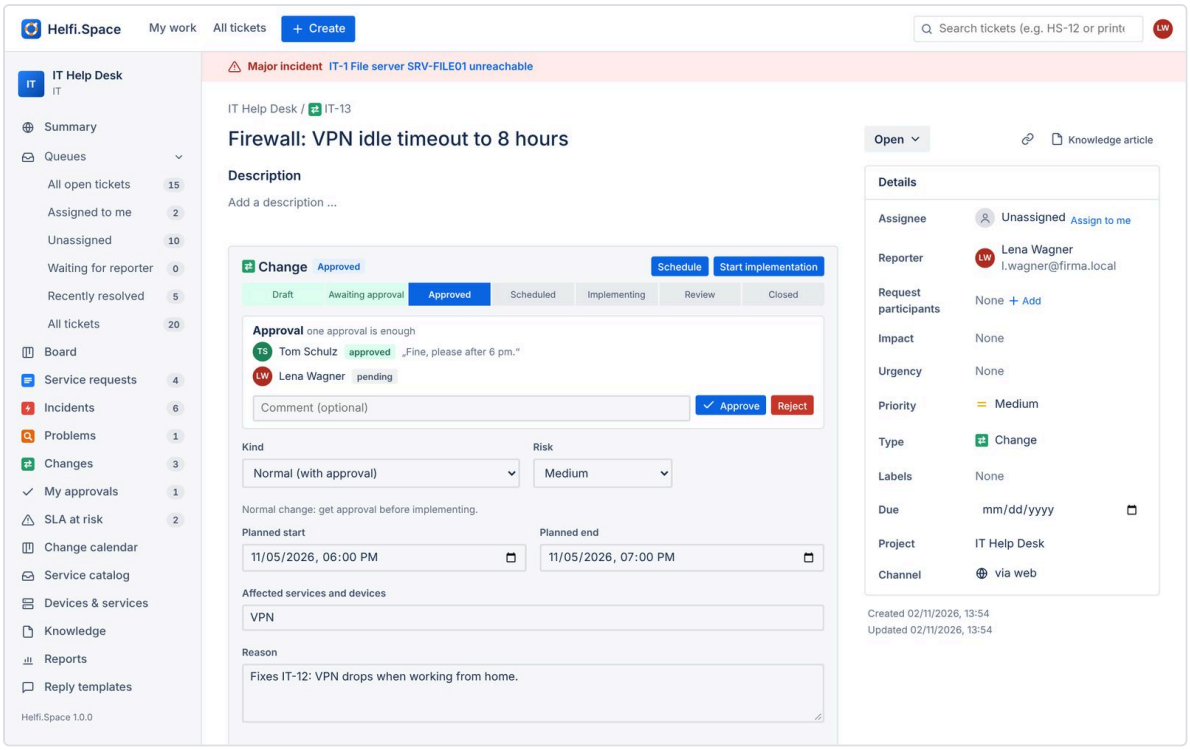
Problem with root cause, workaround and “known error”

4.5 Change management

Planned changes to IT (firewall rule, update, move) are tickets of type **Change**. There are three kinds:

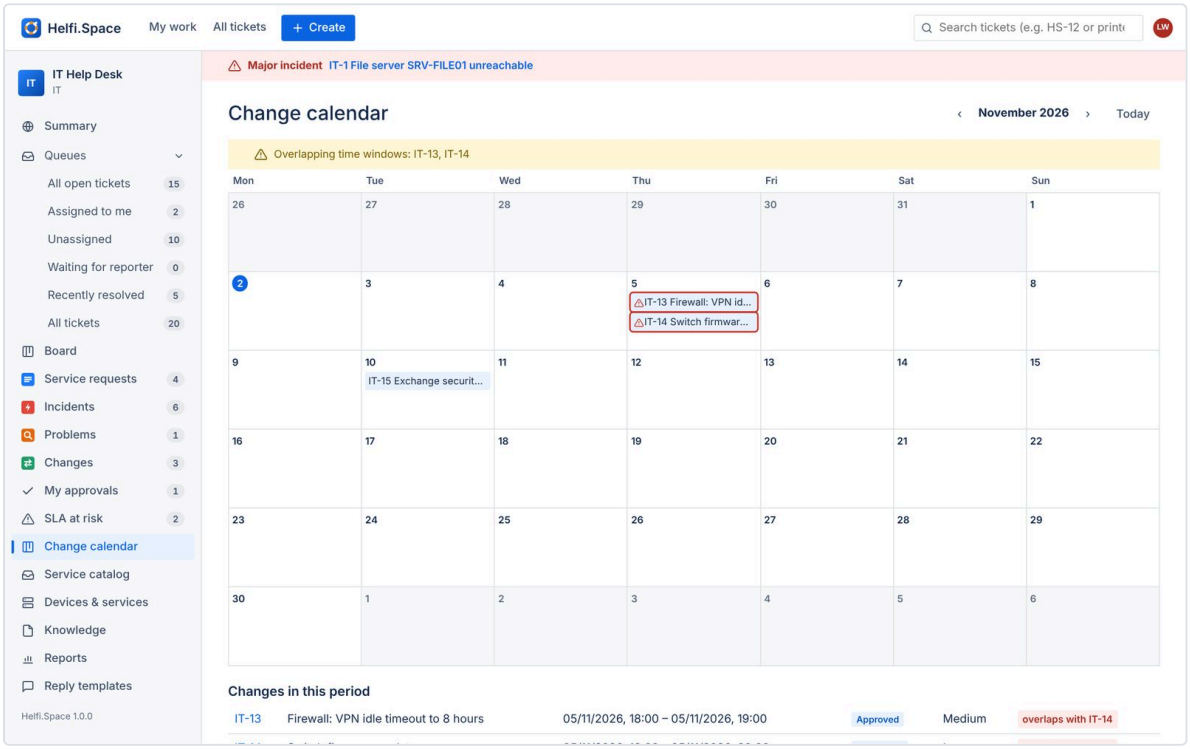
- **Normal:** needs approval.
- **Standard:** pre-approved routine based on a template.
- **Emergency:** implement immediately, approve afterwards.

The *Change* section guides you through the phases Draft → Awaiting approval → Approved → Scheduled → Implementing → Review → Closed. Helfi.Space names missing details (reason, time window, implementation and backout plan) at the respective step. Approvers decide in the ticket or under *My approvals*. When closing you choose the outcome; for “With issues” and “Rolled back” a review is required.



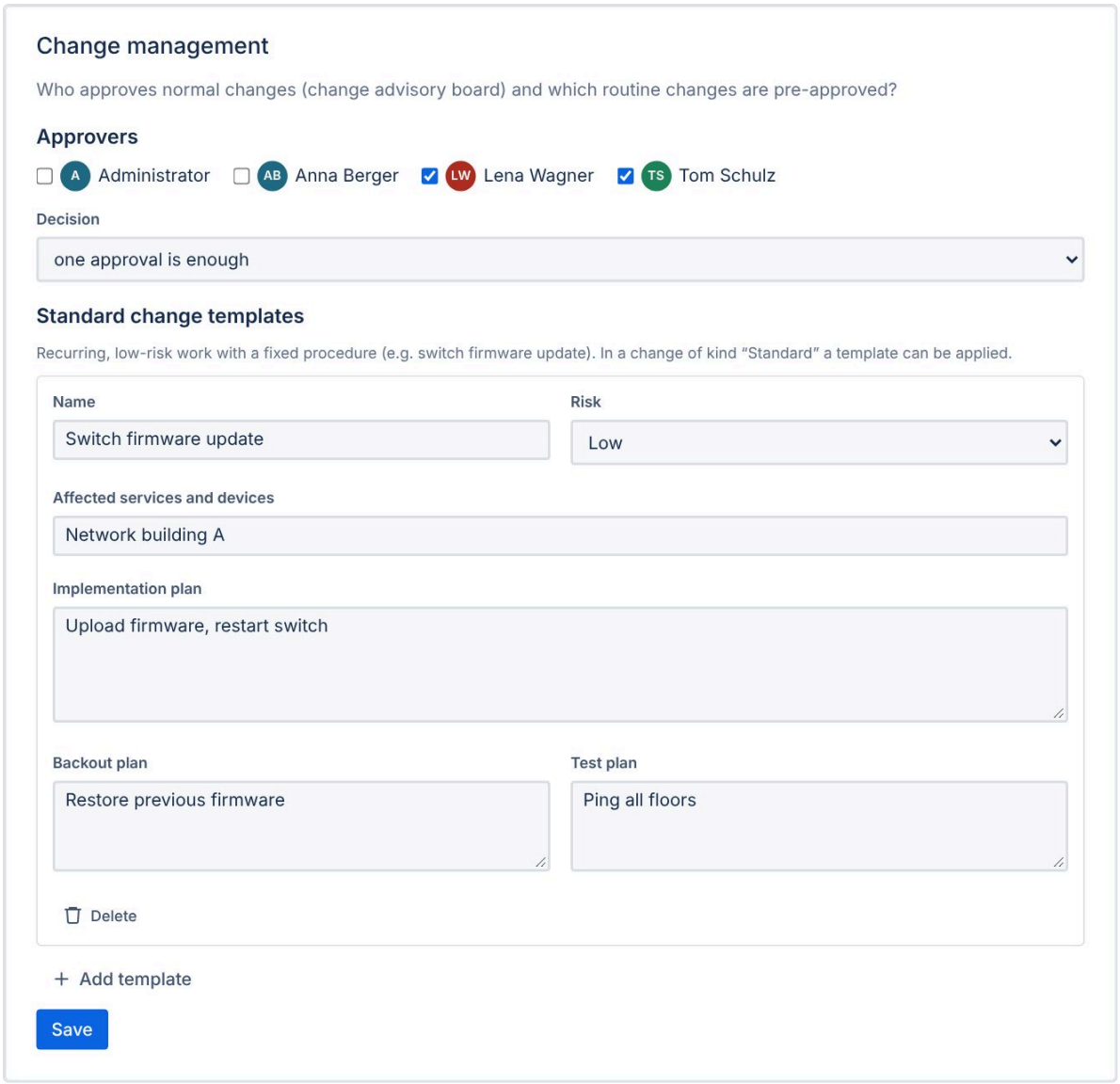
Change with phases, approval and planning

The **change calendar** shows all planned changes in the month; overlaps are marked red.



Change calendar with an overlap

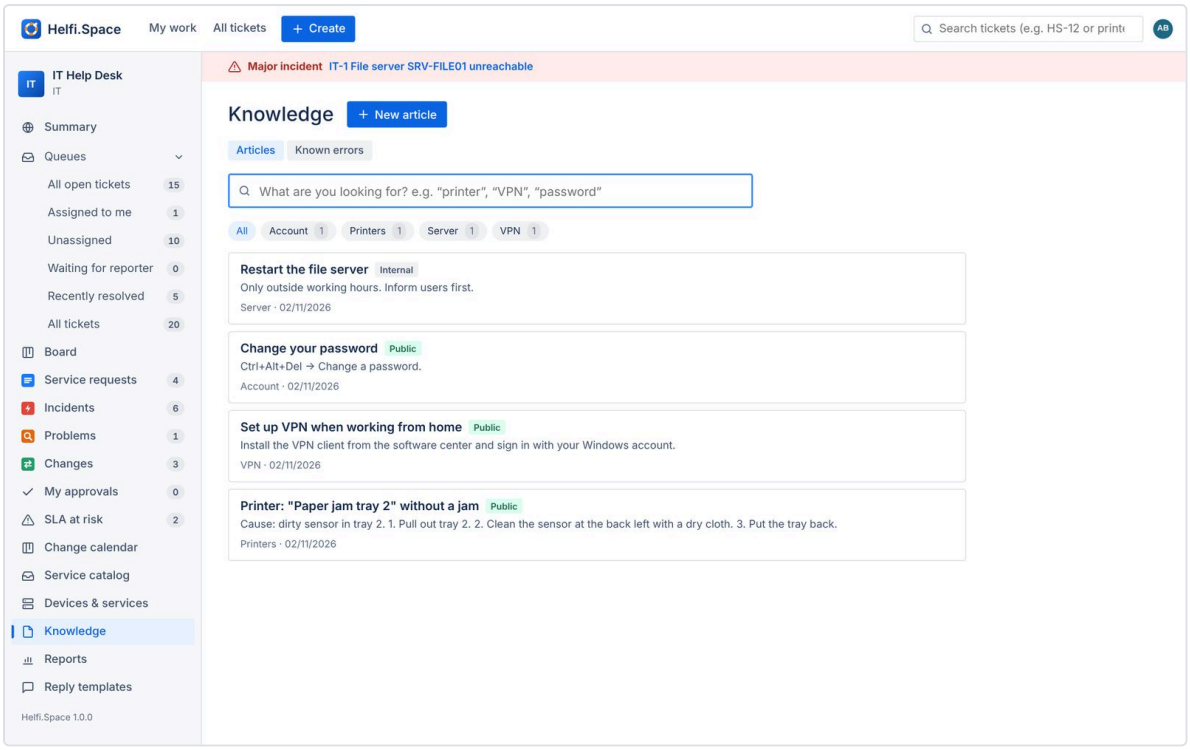
ADMIN Approvers, rule (“one approval is enough” or “everyone must approve”) and templates for standard changes under *Administration → Change*.



Administration → Change

4.6 Knowledge base

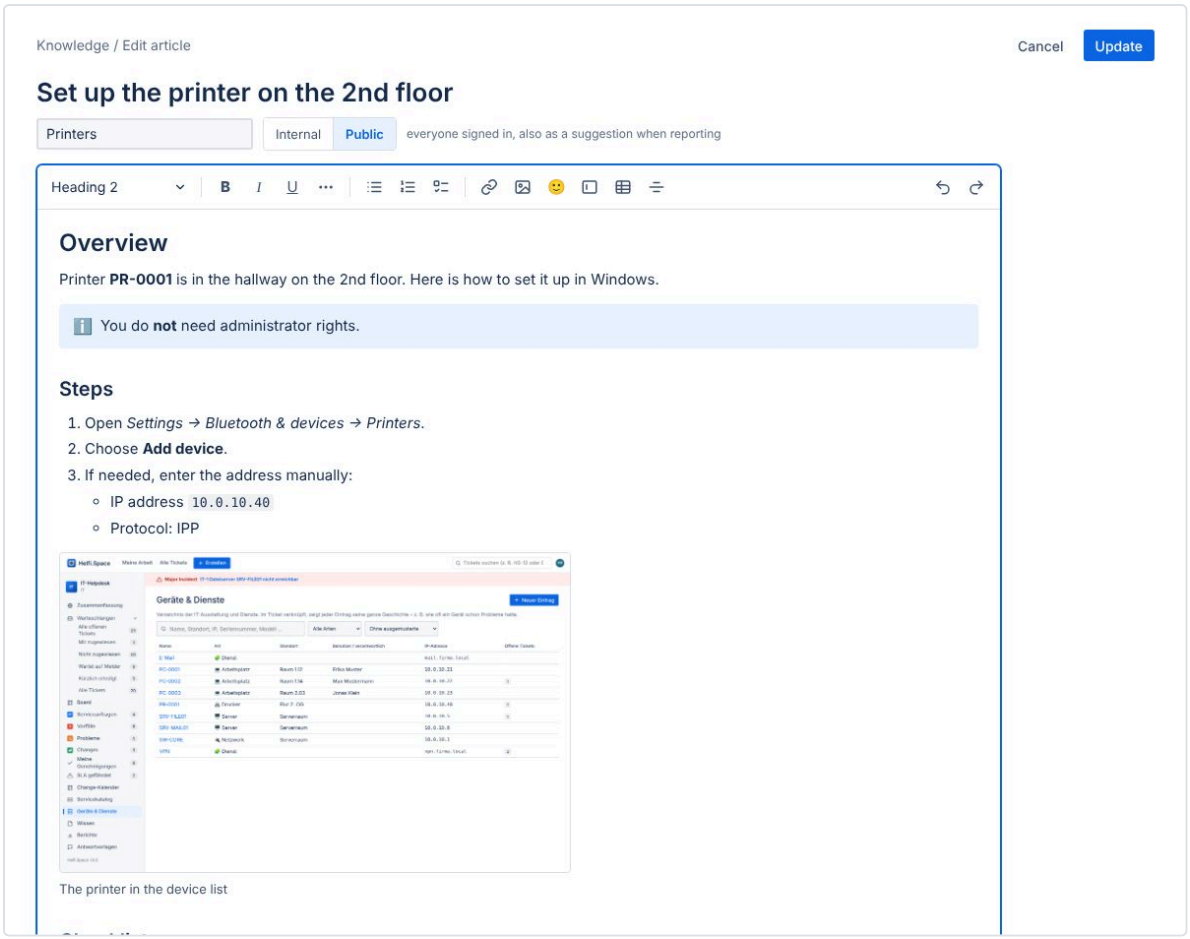
Knowledge collects guides and solutions. Articles are **internal** (agents only) or **public** (everyone signed in, also suggested while reporting). Search ranks title matches higher. From a solved ticket, **Knowledge article** at the top of the ticket creates a draft; remove personal details before publishing. From three headings on, the article view shows a table of contents on the right.



Knowledge base with categories

The editor

You write articles directly in their final look. The toolbar stays at the top while scrolling and offers text style (normal, heading 1–3, quote, code block), bold, italic, underline, under **…** also strikethrough and inline code, bulleted list, numbered list, checklist, link, image, emoji, info panel (note, tip, warning), table and divider.



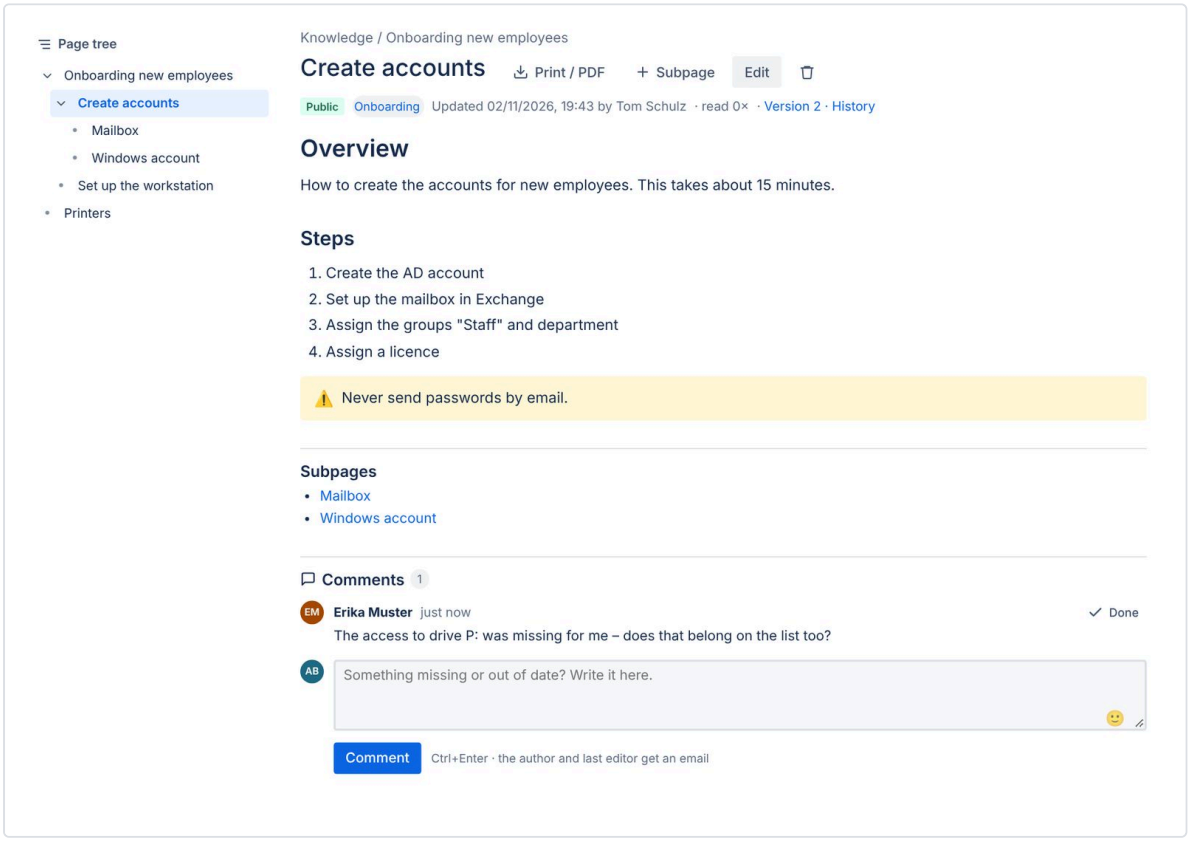
Editor with info panel, nested list and image

- **Images:** with `Ctrl`+`V` from the clipboard, by drag & drop or with the image button. Clicking an image sets its size (small, medium, full width) and a caption.
- **Pasting from Word, web pages or emails:** headings, lists, tables, bold text and links are kept. Images from external web addresses are not taken over; copy them individually as an image.
- **Links** (`Ctrl`+`K`) point to web addresses, email addresses, tickets (`IT-42`) or other articles (`KB-3`).
- **Tables:** `Tab` jumps to the next cell; in the last cell it adds a new row. Add or delete rows and columns with the table button.
- **Draft:** the browser keeps saving unsaved changes. Next time you open the article you can restore them. Leaving the page with unsaved changes asks first.

Type	Result
`#`, `##`, `###` + space	Heading 1–3
`-` or `1.` + space	Bulleted / numbered list (`Tab` indents)
`[]` + space	Checklist (click the box to tick it)
`>` + space	Quote
``` ``` + Enter	Code block (Enter on an empty last line leaves it)
`---` + Enter	Divider
`Ctrl`+`B` / `I` / `U` / `E`	Bold / italic / underline / inline code
`Ctrl`+`Alt`+`1`…`3`, `0`	Heading 1–3, normal text
`Ctrl`+`S`	Save

Page tree

Articles can sit below other articles, for example “Onboarding → Create accounts → Windows account”. In the editor, **Located under** sets where the article sits; on an article, **Subpage** creates an article right below it. On a PC, simply drag articles in the page tree onto another one or onto “top level”. **New folder** in the page tree header creates a short collection page without text of its own, e.g. “Windows” or “Microsoft Office”. The start page shows the page tree and recently changed articles; each article shows its path, the tree on the left and its subpages at the end.

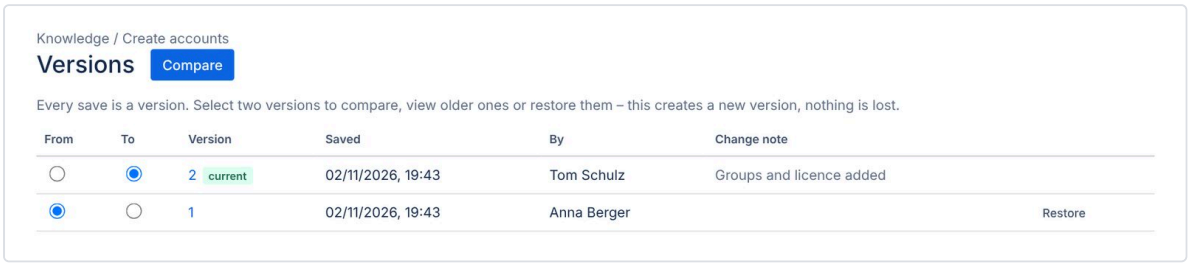


Article with page tree, subpages and a comment

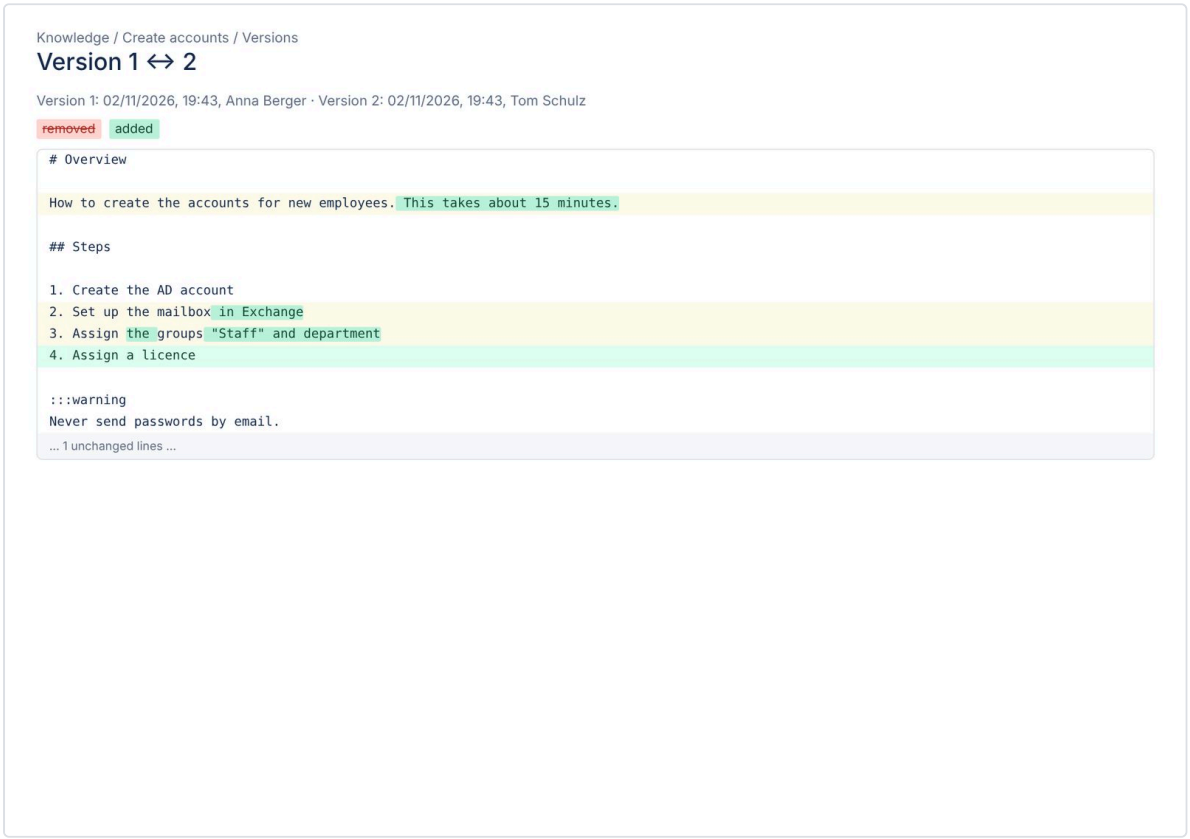
- Users only see public pages. If a public page sits below an internal one, it moves up a level for users.
- If you delete an article with subpages, they move up one level. Moving creates no new version.

Versions

Every save with a change is a version. In the editor header you can add a short **change note**. *Version n · History* on the article lists all versions. Select two and **compare** them: removed text is red and struck through, new text green. You can view older versions and **restore** them; this creates a new version, the history stays complete.



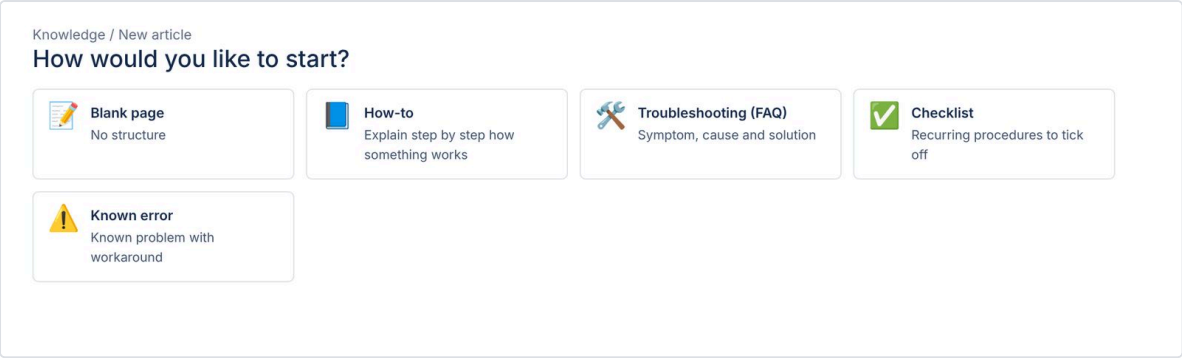
Versions of an article



Comparing two versions

Templates

New article first asks how you want to start: blank page, *How-to*, *Troubleshooting (FAQ)*, *Checklist*, *Known error* or a template of your own. Save your own templates in the editor with the copy icon (*Save as template*) and manage them under *Knowledge → Templates*.



Choosing a template for a new article

Comments

Anyone who can see an article can comment below it – including users on public articles. The author and last editor get an email. Agents mark comments as **done**; done comments collapse. You can delete your own comments. In the article list, 💬 shows the number of open comments.

Print and PDF

Print / PDF on the article opens the print dialog; choose “Save as PDF” there. Only title, path, date and content are printed, without page breaks inside images, tables or info panels. If the article has subpages you can print **with all subpages**: you get one document with a table of contents, each page starting on a new sheet – for example a complete onboarding handbook.

4.7 Setting up the service catalog ADMIN

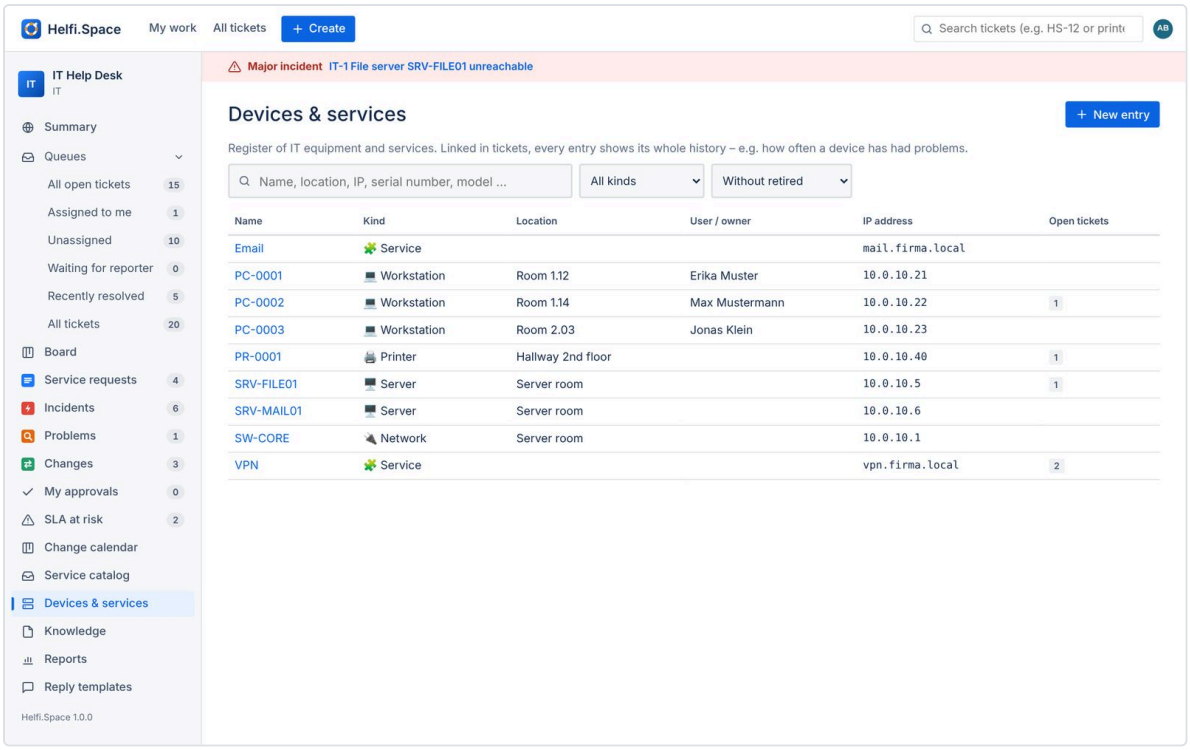
Under *Administration → Service catalog* you create offers: name, emoji, description, project and form fields (text, multi-line, choice, check box, date; each optionally required). Optionally choose approvers, also without the agent role, and whether one or all approvals are needed. Requests become tickets of type *Request*; the answers appear in the description. How users see the catalog is described in 2.9.



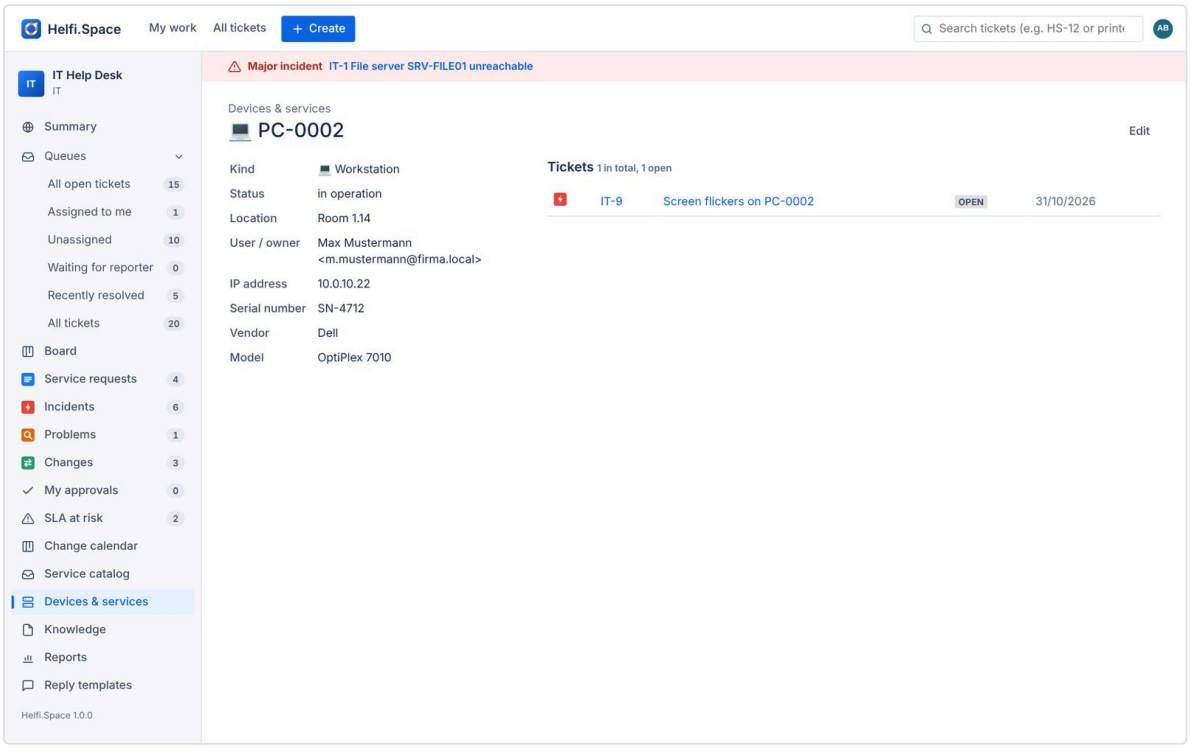
Administration → Service catalog

4.8 Devices & services

A lean register of IT equipment and services (CI register). Each entry has a kind, status (in operation, in maintenance, retired), location, user, IP address, serial number, vendor and model. In a ticket you link affected devices and services in the *Devices & services* section. For changes these are the affected services. An entry’s detail page lists all its tickets, so recurring faults stand out.



Devices & services with open tickets per entry

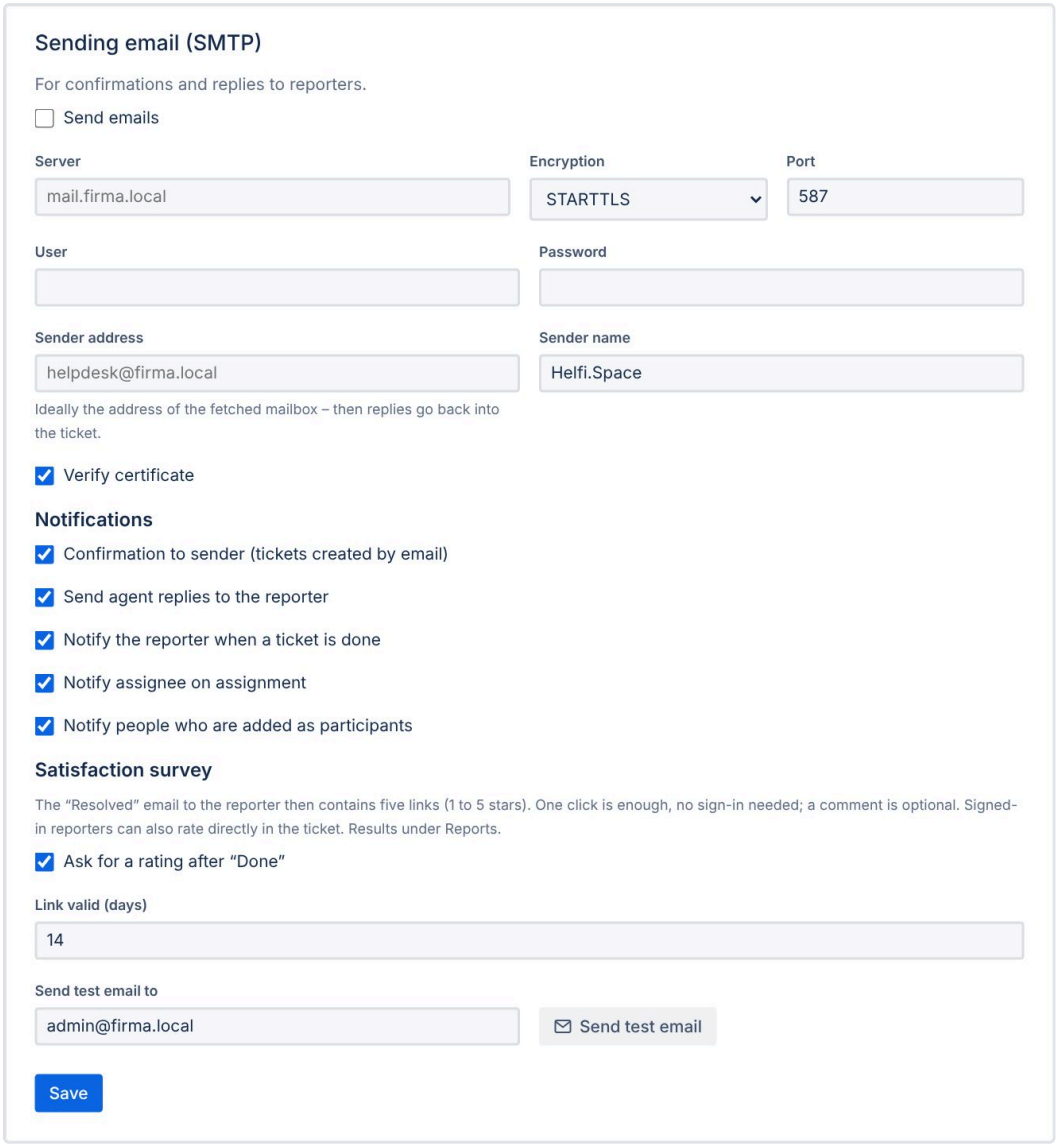


A device with its ticket history

ADMIN **CSV import** from your inventory tool: the first line contains the column names (name/hostname, type, location, IP address, serial number, vendor, model, user, status – German or English). Values like “Notebook” or “Switch” are mapped to the matching kind. Existing names are updated. Only administrators can delete an entry, and only while no tickets are linked; otherwise set it to “retired”.

4.9 Satisfaction survey ADMIN

Switch it on under *Administration → Email → Satisfaction survey*. The “Resolved” email to the reporter then contains the rating links (2.12). Each link contains a random token; only its hash is stored. The link is valid for a configurable number of days and only shows the ticket key and title. If a ticket is reopened and resolved again, there is a new survey. Agents see the rating at the top of the ticket.



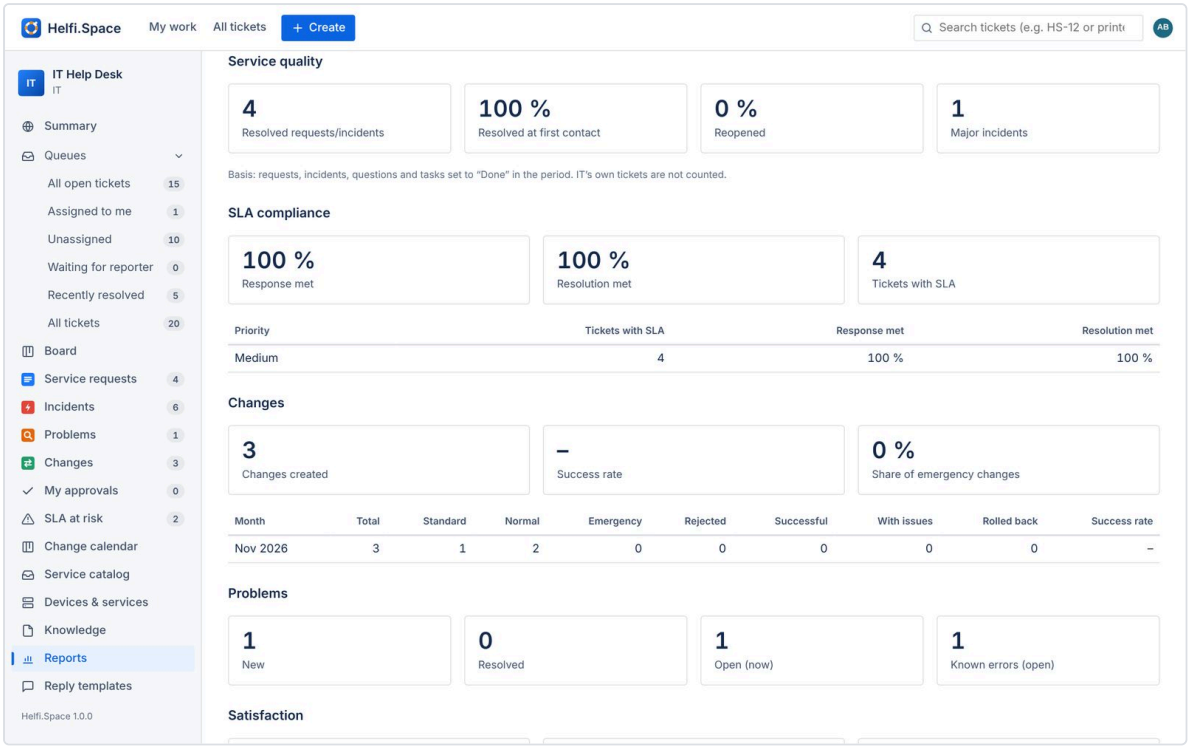
Administration → Email with the satisfaction survey

4.10 ITIL key figures

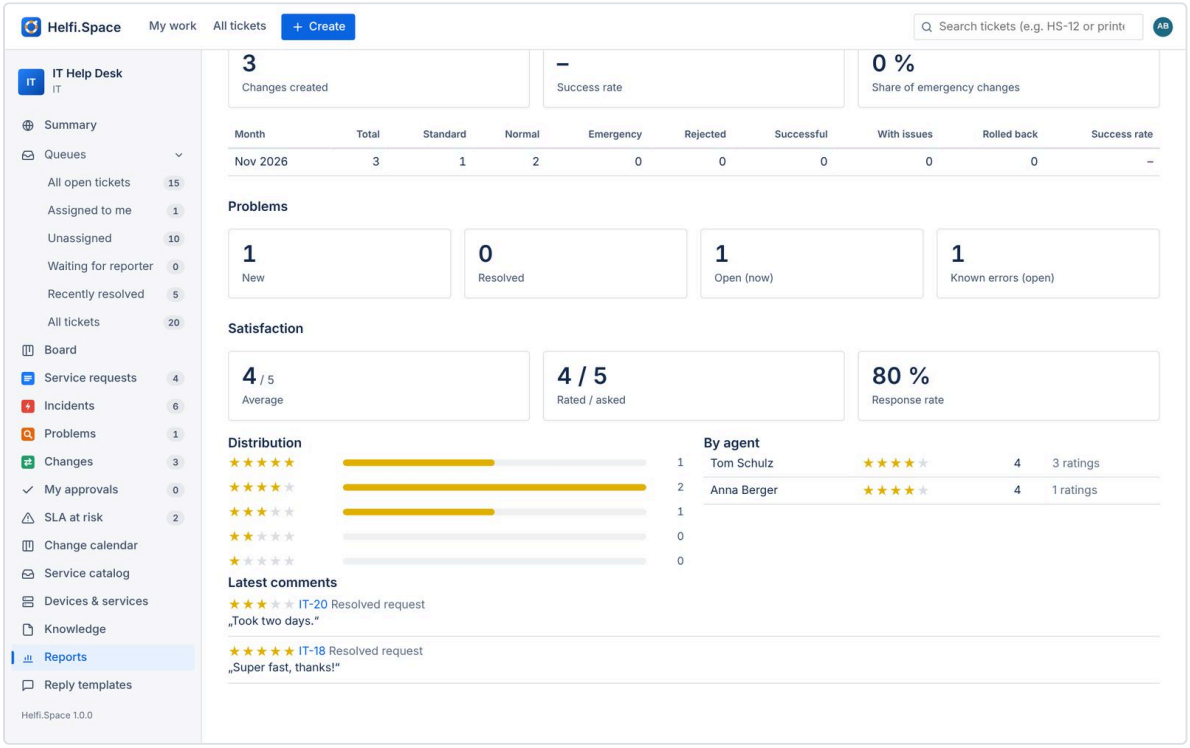
Under *Reports*, the agent statistics (3.9) are followed by further figures for the same period:

- **Service quality:** resolved requests and incidents, resolved at first contact (at most one reply, no follow-up question, never reopened), reopen rate, major incidents.
- **SLA compliance** for response and resolution, overall and per priority.
- **Changes** per month: kinds, rejected, outcomes, success rate and share of emergency changes.
- **Problems:** new, resolved, open, known errors.

- **Satisfaction:** average, response rate, distribution of stars, values per agent and the latest comments.



ITIL key figures in the report



Satisfaction in the report

5 For administrators: setup ADMIN

5.1 Installation

You need Docker with Compose (Linux server, Synology …). Download `docker-compose.yml`, `env.example` and `update.sh` from the latest release, then:

```
mkdir helfi && cd helfi
cp env.example .env
nano .env                 # adjust at least HELFI_BASE_URL
docker compose up -d
docker compose logs helfi # shows the setup code
```

Afterwards only four things live on the server:

`docker-compose.yml`	usually stays the same across updates
`.env`	your settings (address, port, version, PUID/PGID)
`update.sh`	manual update (optional)
`data/`	database, attachments, logo, backups, key file

Important entries in `.env`:

`HELFI_BASE_URL`	address under which Helfi.Space is reachable (for links in emails and SSO)
`HELFI_HTTP_PORT`	port on the server (default 8080)
`HELFI_VERSION`	`latest` or a fixed version, e.g. `1.5.0`
`PUID` / `PGID`	owner of the files in `data/` (Synology e.g. `1026` / `100`; not 0)
`HELFI_TRUST_PROXY`	address of the reverse proxy (see next section)
`HELFI_SECRET`	optional: own key for stored passwords – set it only during setup
`TZ`	time zone, e.g. `Europe/Berlin`

5.2 HTTPS and reverse proxy

Helfi.Space itself speaks HTTP on port 8080. In production put a reverse proxy with HTTPS in front of it (Synology reverse proxy, Nginx Proxy Manager, Caddy, Traefik …). Then set `HELFI_BASE_URL=https://helfi.firma.local` – cookies are marked `Secure` automatically.

```
helfi.firma.local {
    reverse_proxy 10.0.10.20:8080
}
```

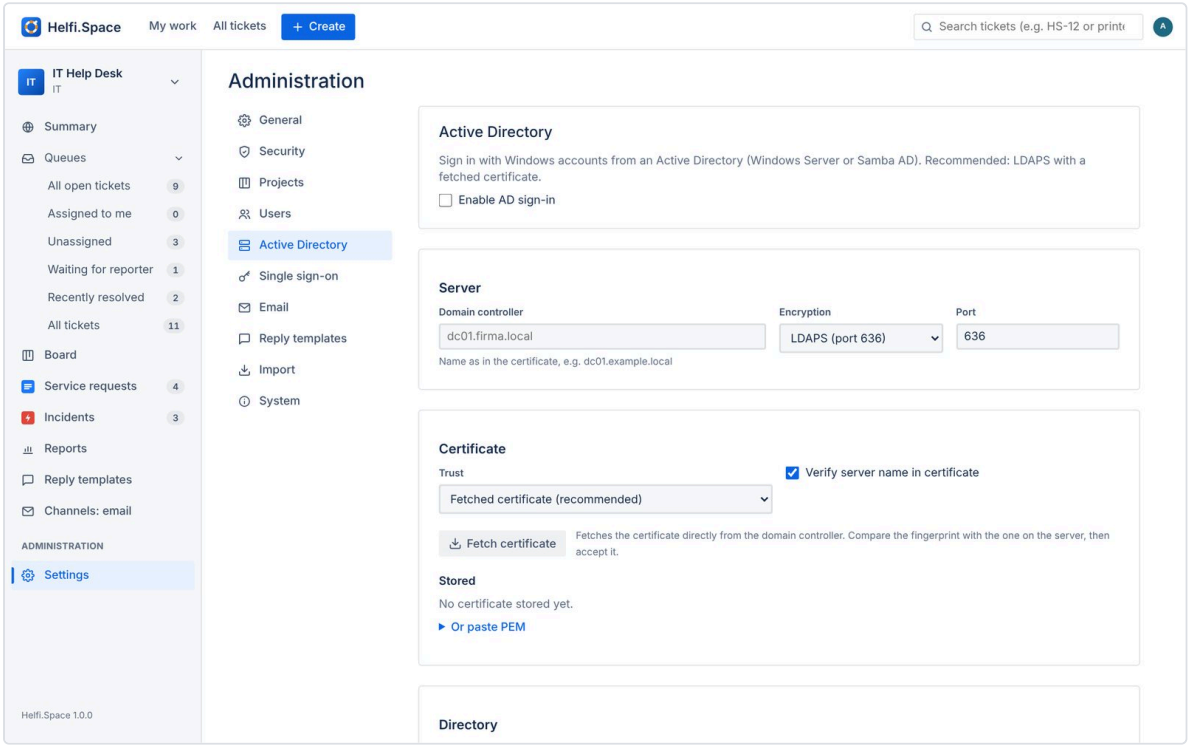
> Enter the proxy's address in `HELFI_TRUST_PROXY`, e.g. `10.0.10.5` (several separated by commas: `loopback, 10.0.10.5`). Without it Helfi.Space only trusts `X-Forwarded-For` from its own host (`loopback`); values such as `true` or a number are ignored with a warning in the log. If the proxy runs on another host, Helfi.Space then only sees the proxy: the brake against password guessing then hits everyone together, and every request counts as "from outside" to be safe.

5.3 Initial setup

On the first visit to `HELFI_BASE_URL` the setup wizard starts. It asks for the **setup code** from the container log (`docker compose logs helfi`) and creates the **local administrator** and the first project. It then continues straight to the AD setup.

5.4 Active Directory

Administration → Active Directory



Active Directory settings

1. Enter the **domain controller** with the name in its certificate (e.g. `dc01.firma.local`). Encryption: LDAPS (636) or StartTLS (389).
2. **Fetch certificate**, compare the SHA-256 fingerprint with the one on the server, then **Accept certificate**.
   Windows: `certlm.msc` → Personal → certificate of the DC. Samba: `openssl x509 -in /var/lib/samba/private/tls/cert.pem -noout -fingerprint -sha256`
3. Enter a **service account**, e.g. `svc-helfi@firma.local` – a normal domain account without special rights.

4. **Groups** for administrators and agents, e.g. `Helfi-Admins` and `Helfi-Agents` (name or DN, nested groups count). Optionally a group of “Allowed users”.
5. **Test connection**, ideally with a test sign-in. An empty base DN is filled in; a Microsoft certificate authority (AD CS) that is found can be accepted too, so renewed DC certificates stay valid.
6. **Enable AD sign-in** and save.

Users are created at their first sign-in or when an agent adds them as a participant from AD (**Suggest participants from AD**, on by default, requires the service account). Their role follows the groups at every sign-in unless an admin locked it under *Users* with “Role locked”. Helfi.Space checks daily whether the certificate still matches and whether it expires soon, and shows problems as a notice.

5.5 Single sign-on

Administration → Single sign-on (OpenID Connect: Microsoft Entra ID, ADFS, Keycloak, Authentik …)

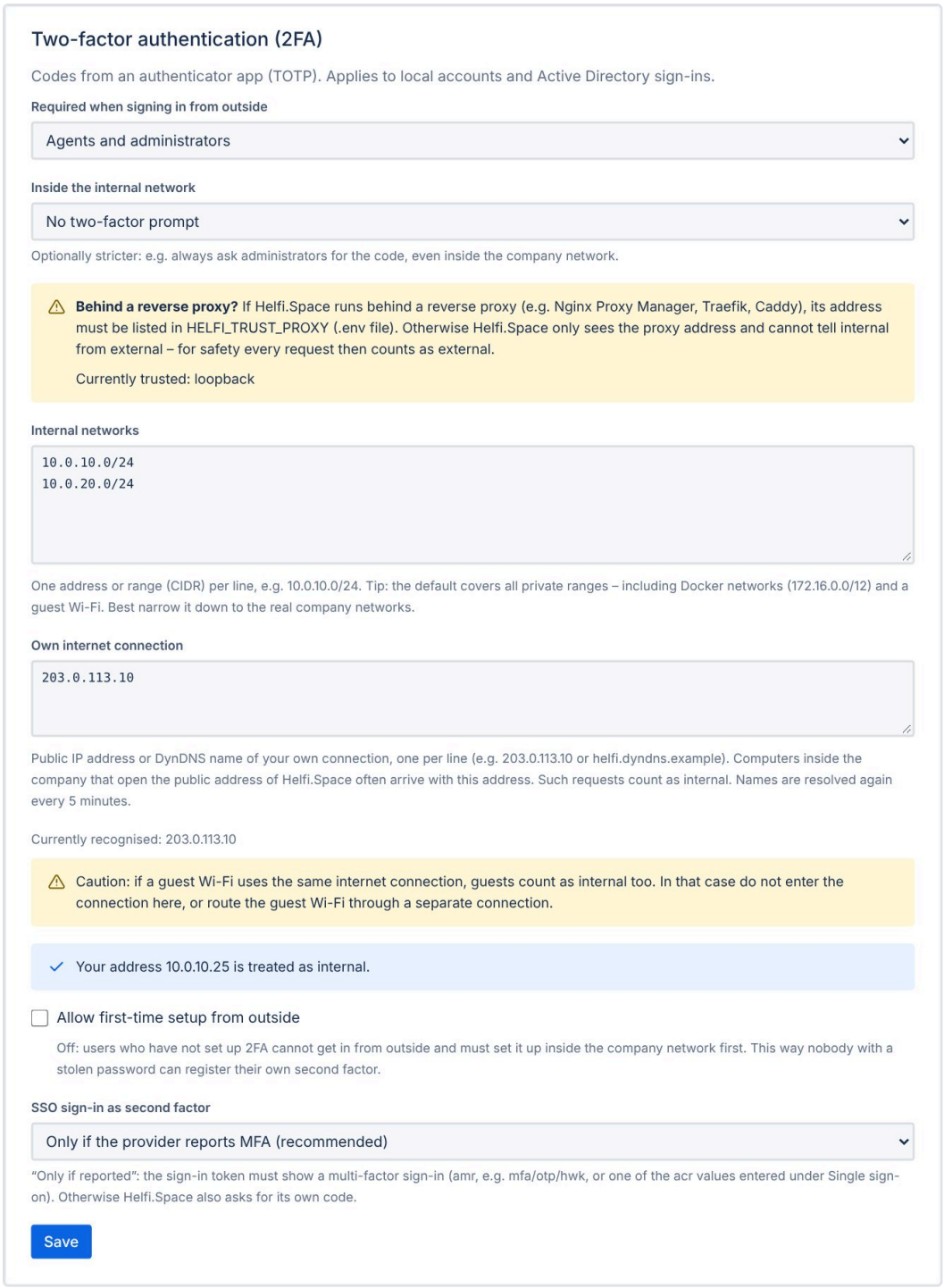
1. Create an app at the provider and enter the redirect URI shown, e.g. `https://helfi.firma.local/auth/oidc/callback`.
2. Enter issuer URL, client ID and client secret and click **Test**. Entra ID: `https://login.microsoftonline.com/<tenant ID>/v2.0` and add the groups claim under *Token configuration*.
3. Enter the admin and agent groups (group IDs for Entra ID).
4. Optionally “redirect straight to SSO”. The normal sign-in remains available at `/login?local=1`.

SSO and AD together: with “Merge with AD accounts for domain” (e.g. `firma.local`), `m.muster@firma.local` from SSO is the same account as the AD user `m.muster`. Only enter it if the provider does not allow guest accounts from other organisations. If the token contains only the email address (no `preferred_username`/UPN), accounts are only merged if the provider reports it as verified (`email_verified`). Email addresses from SSO are only taken over if they are verified or equal the user name of the allowed domain.

Stable via the AD identifier (recommended): under *Claim with AD identifier* enter the claim holding `objectSid` or `objectGUID` – `onprem_sid` (optional claim) for Entra ID, a custom mapping for Keycloak/Authentik. Then accounts are merged only this way, not by freely chosen names. With a service account the AD account is checked on every SSO sign-in (disabled, group, role); if the AD is unreachable, the SSO sign-in is cancelled.

5.6 Two-factor authentication and networks

Administration → Security



Two-factor requirement, internal networks and own internet connection

Setting	Options
Required when signing in from outside	all users (default) · agents and administrators · administrators only · nobody
Inside the internal network	no prompt (default) · administrators only · agents and administrators · all users

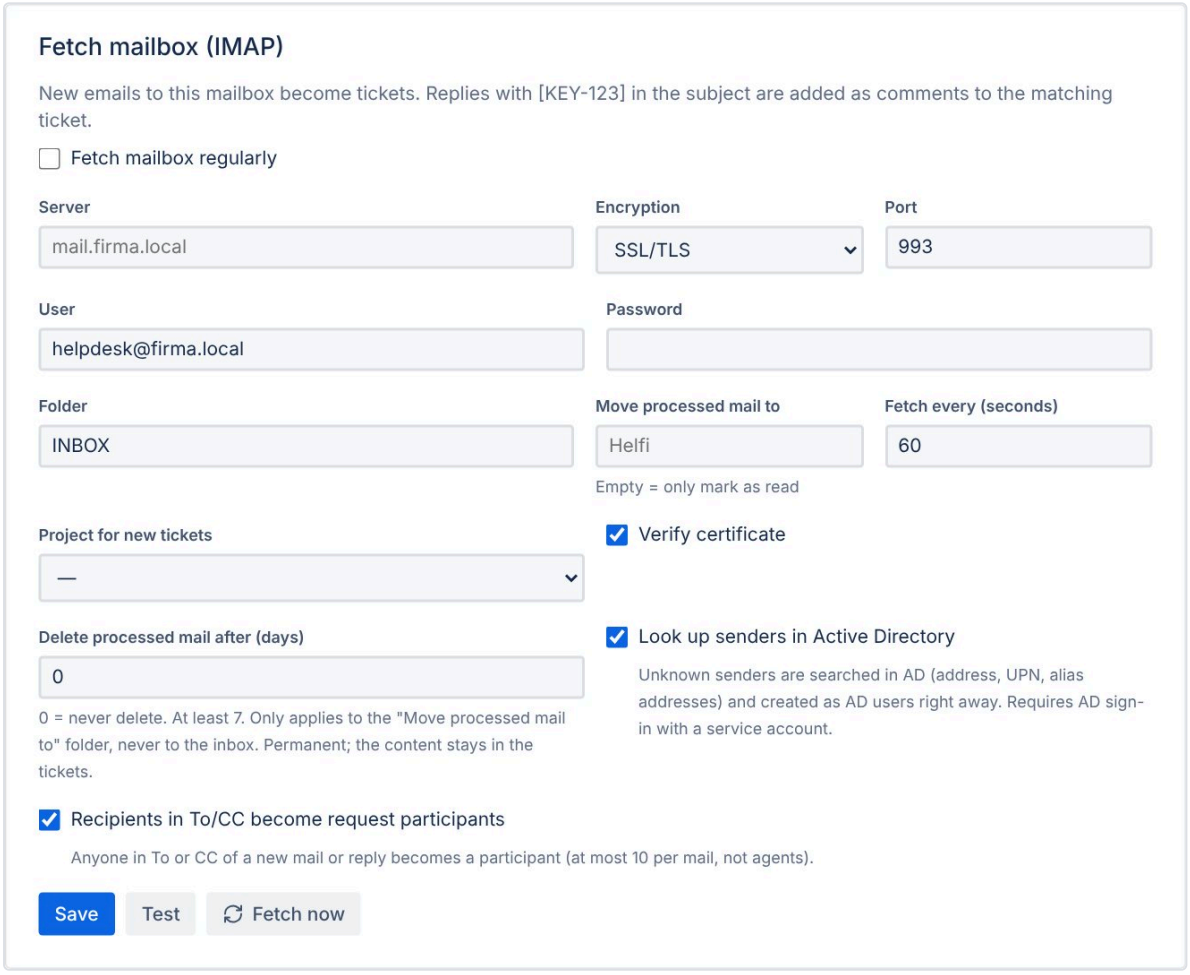
- **Internal networks:** one address or range per line. The default covers all private ranges, including Docker networks and a guest Wi-Fi – best narrow it down to the real company networks, e.g.

`10.0.10.0/24`.

- **Own internet connection:** public IP or DynDNS name (e.g. `203.0.113.10` or `helfi.dyndns.example`). Requests from there count as internal; names are resolved again every 5 minutes. Careful with a guest Wi-Fi on the same connection.
- **Reverse proxy:** without a matching `HELFI_TRUST_PROXY` every request counts as external; the page shows the current value.
- **Allow first-time setup from outside:** off by default.
- **SSO sign-in as second factor:** *Only if the provider reports MFA* (default: the token shows a multi-factor sign-in via `amr` or an entered `acr` value), *Always* or *Never*. Otherwise Helfi.Space asks for its own code after the SSO sign-in. Installations with SSO enabled keep "Always" on update.
- **Reset 2FA** for a person: *Users → Reset 2FA*.

5.7 Email

Administration → Email



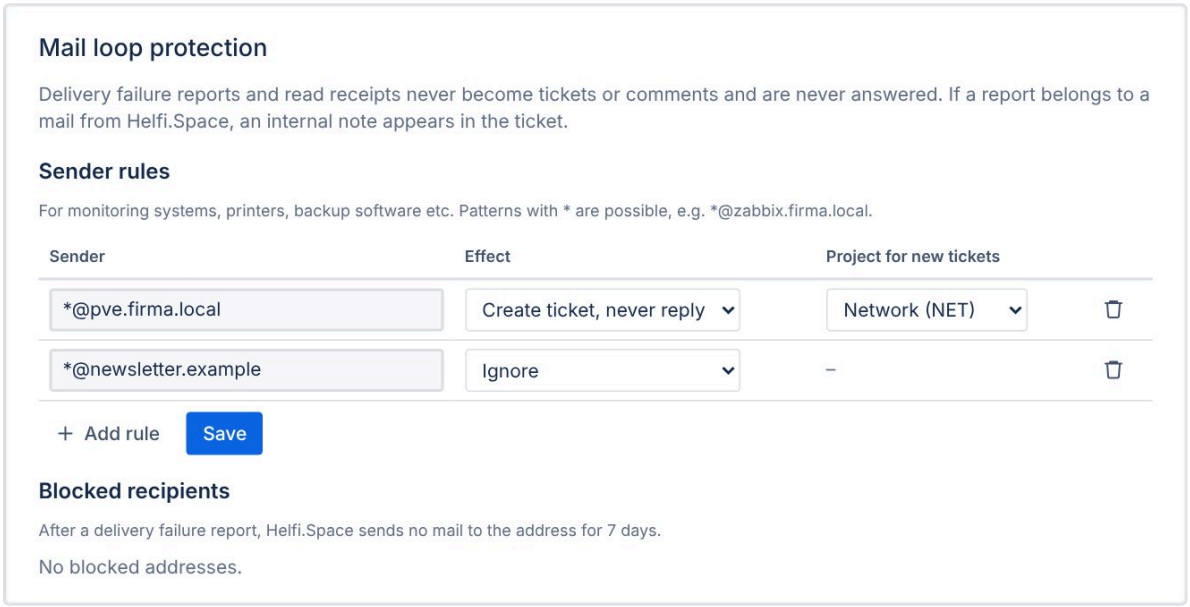
Fetching a mailbox (IMAP)

- **Fetch mailbox (IMAP):** server, user and password of the help desk mailbox. New emails become tickets, replies become comments. Processed emails are marked as read or moved to a folder – a separate folder is recommended. Skipped emails and delivery failure reports stay in the inbox, marked as read, and are not deleted automatically. Processed emails get their own flag; if the

mailbox is read in Outlook in parallel, they are still not lost (if the mail server supports custom flags, otherwise better not read it in parallel).

- **Replies by ticket key** in the subject are only accepted from the reporter, request participants and everyone who has already written in the ticket; anyone who merely guesses the key gets a new ticket. Replies to mails from Helfi.Space are matched via the Message-ID.
- **Sending (SMTP):** ideally use the mailbox address as sender, so replies come back into the ticket.
- **The sender is always the ticket system** – never an agent's address, no personal signatures. So the reporter knows who looks after them, reply and resolved mails end with "Kind regards", the agent's name and the system name; if the ticket is assigned to someone else, "Your contact person: …" is added. In the browser, users see the field as "Your contact".
- **Look up senders in Active Directory:** unknown senders are searched in AD and created as AD users right away. Otherwise they become *email contacts*; when they sign in later, their tickets are theirs.
- **Recipients in To/CC become participants** (can be turned off) – for new tickets and replies to mails from Helfi.Space, not when a reply is matched only by the key in the subject.
- **Retention:** "Delete processed emails after … days" (default: never). Only the folder for processed emails is cleaned, never the inbox.

Mail loop protection and sender rules



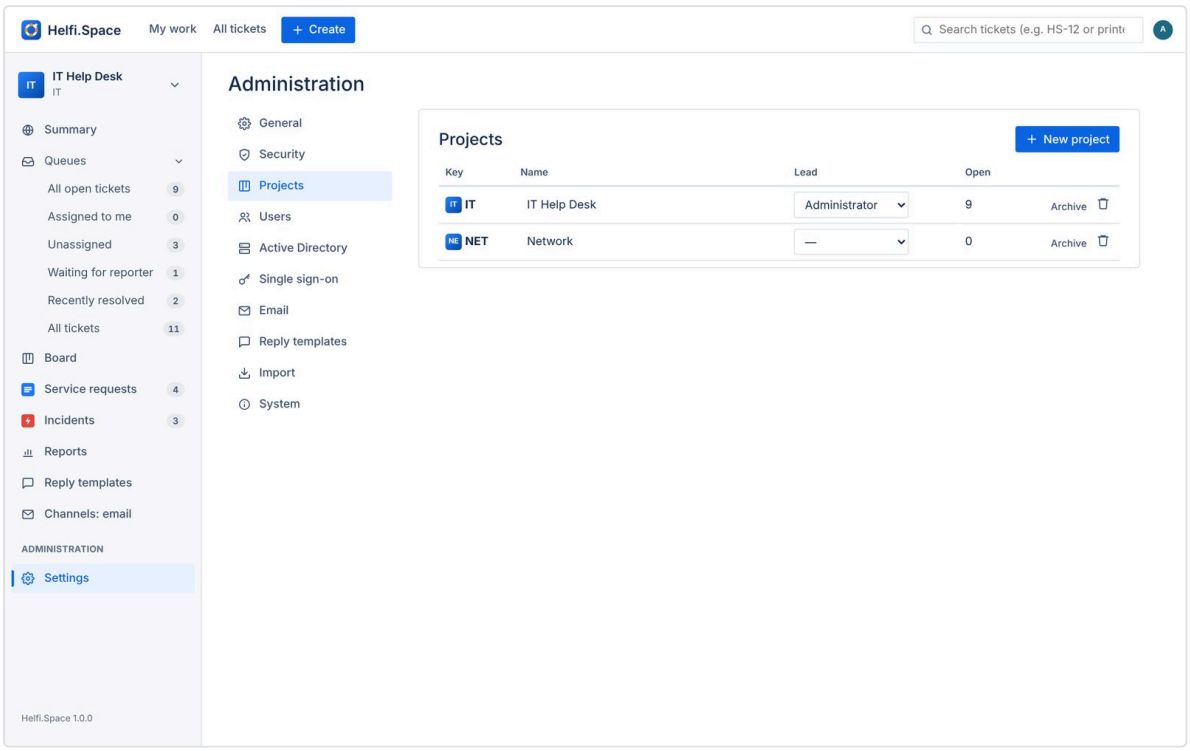
Sender rules and blocked recipients

- Delivery failure reports and read receipts never become tickets and are never answered. The affected address gets no emails for 7 days (*Blocked recipients*, can be released early there). Only addresses from real header lines of the report or from its `message/delivery-status` part are blocked.
- **Sender rules** (address or pattern like `*@pve.firma.local`): "Create ticket, never reply" – optionally into a separate project – or "Ignore".
- **Automatically generated emails** (Proxmox, NAS, monitoring; recognised by `Auto-Submitted`, `Precedence: bulk` or `noreply@`) are **only accepted with the rule "Create ticket, never reply"**. *Recently skipped emails* lists sender and reason.

- More than 20 new tickets per hour from the same sender are ignored.

5.8 Projects

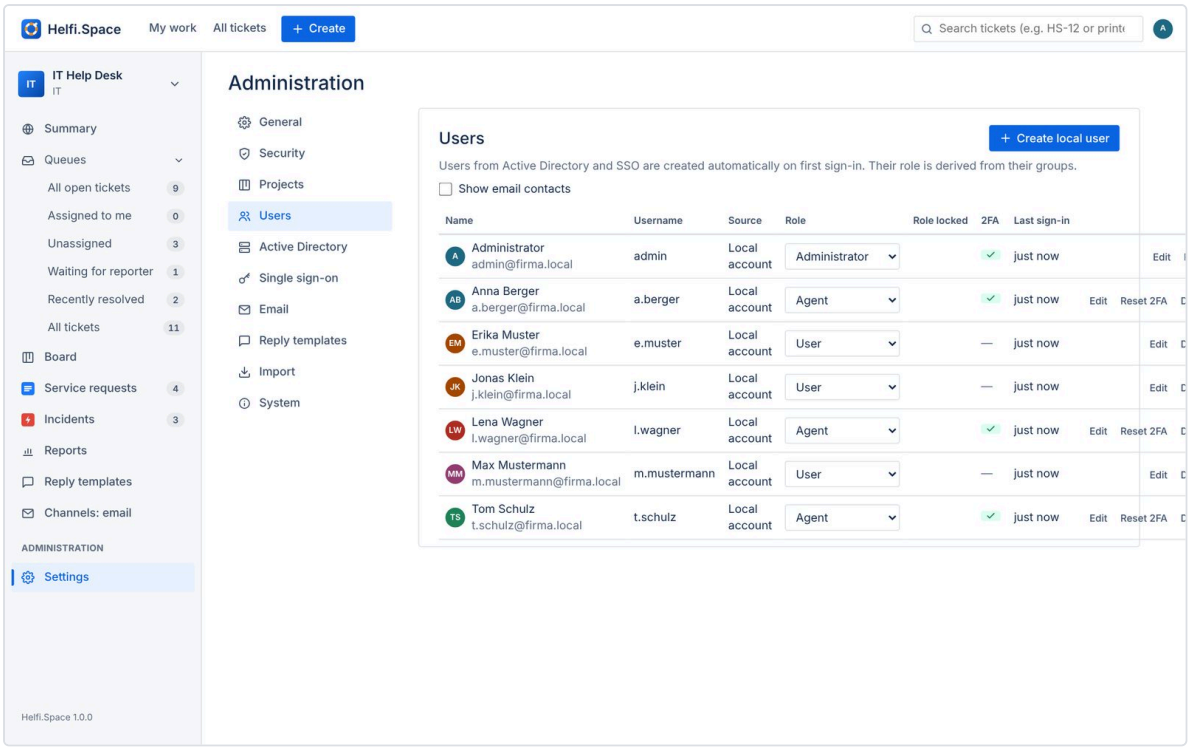
Administration → Projects: create projects (key such as `IT` or `NET` and a name), set the lead, **archive** (kept with all tickets, disappears from the selection) or delete. A project with tickets can only be deleted if its tickets are moved to another project; a backup is made automatically beforehand.



Projects

5.9 Users

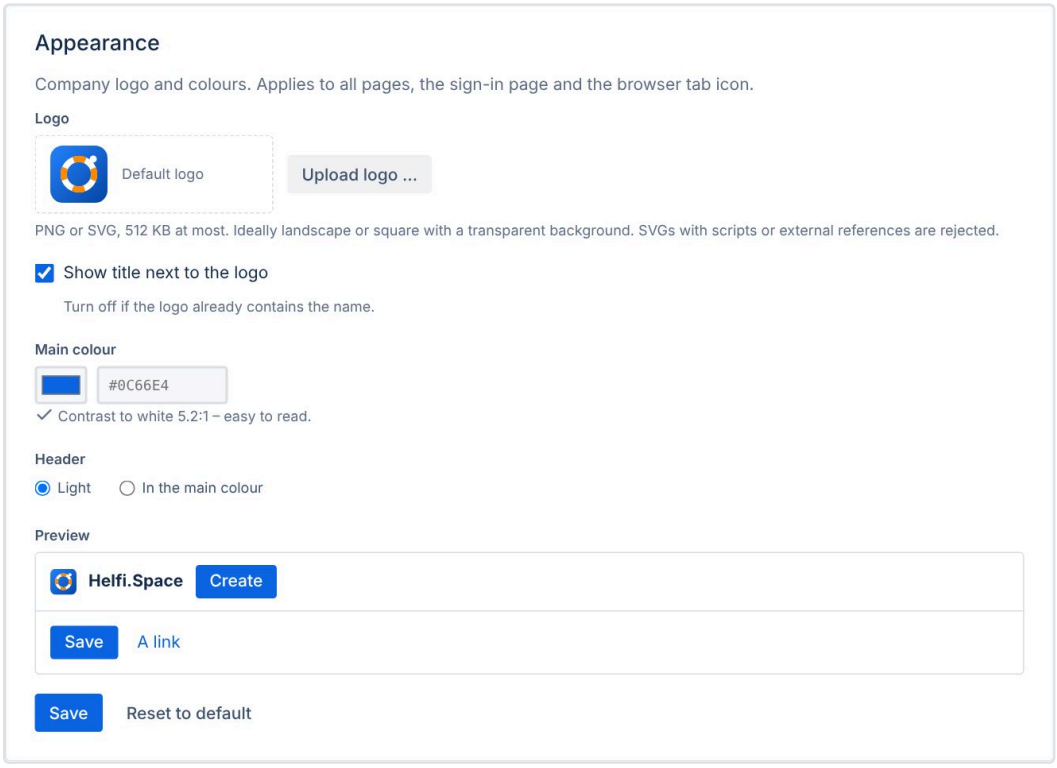
Administration → Users lists all accounts with origin (local, AD, SSO, email contact), role, 2FA status and last sign-in. Here you create local users, change roles (“Role locked” prevents AD groups from overriding them), deactivate accounts and reset 2FA.



User management

5.10 Appearance

Administration → General → Appearance



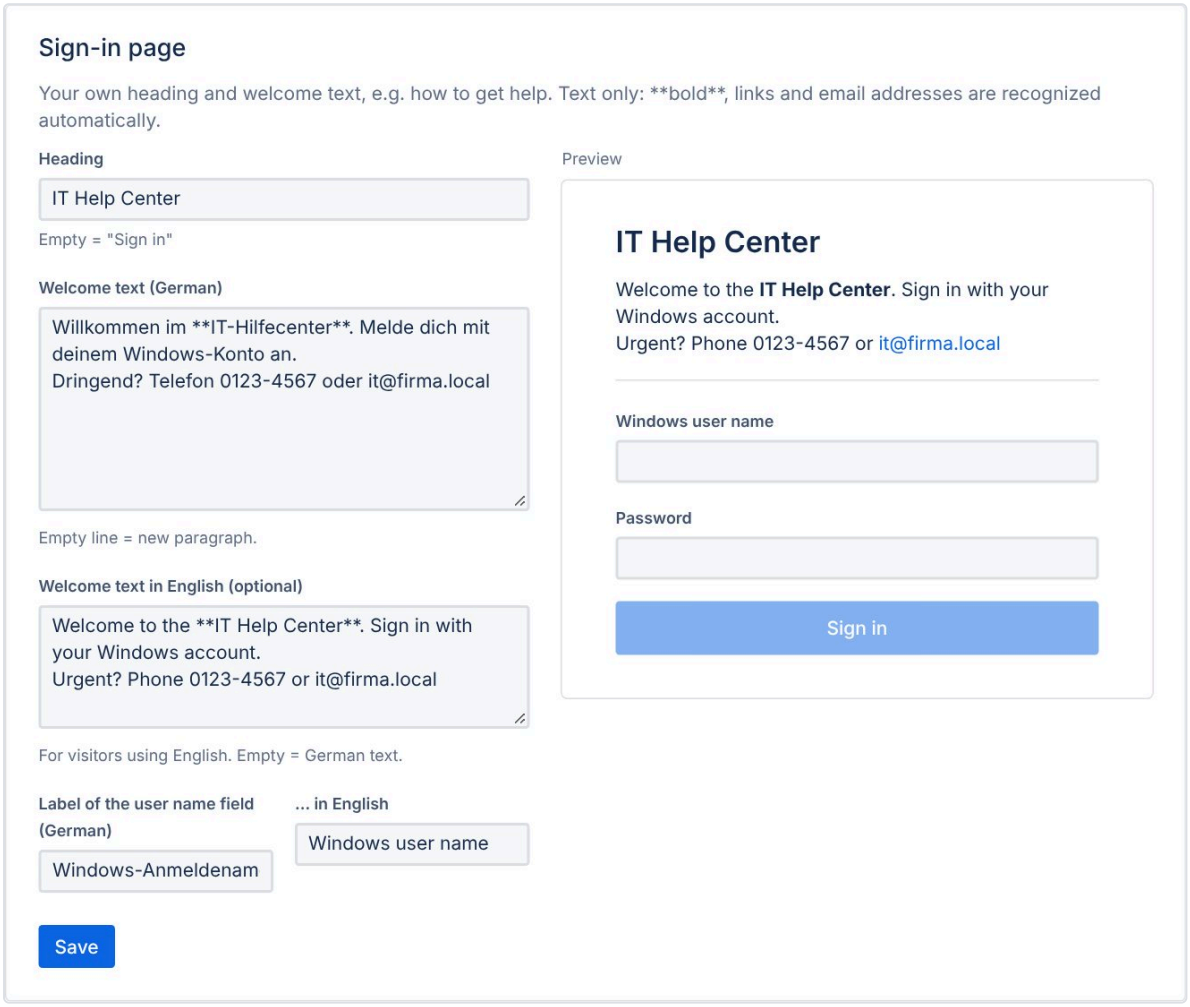
Logo, main colour and header with preview

- **Logo** as PNG or SVG (512 KB at most) for header, sign-in page and browser tab. SVGs with scripts or external references are rejected.

- **Main colour** with contrast check (warning below 4.5:1) and **header** light or in the main colour.
- *Reset to default* restores everything.

5.11 Sign-in page

Administration → General → Sign-in page: your own heading, a welcome text in German and English (e.g. how to get help in an emergency) and your own label for the user name field ("Windows user name"). Text only: `**bold**`, links and email addresses become clickable, HTML is never executed.



Customising the sign-in page with preview

5.12 Import from another ticket system

Administration → Import takes over projects, tickets, comments and attachments via the REST API of the previous system (Cloud and Server/Data Center). Credentials are not stored.

1. **Connect** (address, email and API token, or personal access token).
2. **Choose projects** and the target project.
3. **Map statuses and types** (suggestions are pre-filled).
4. **Import** – with progress, can be cancelled.

Ticket keys are kept (`SUP-123` stays `SUP-123`), people are matched by email address, and no emails are sent. Before every import starts, Helfi.Space creates a backup. A second run only updates what

changed and never creates duplicates – so you can import once as a trial and again on the cut-over day.

6 For administrators: operation ADMIN

6.1 Updates

With container management (recommended)

Dockhand, Watchtower or Tugtainer detect new images of `ghcr.io/soleil0815/helfi.space.ticket:latest` and update the container – ideally at night. Helfi.Space backs up the database itself when a new version starts (`data/backups/helfi-vor-<version>-….sqlite`) and then migrates it.

Manually

```
./update.sh             # latest version
./update.sh 1.5.0       # specific version (recorded in .env)
./update.sh -y          # without questions
./update.sh --cleanup   # show leftovers of earlier updates and delete them after
confirmation
./update.sh --offline helfi-1.5.0-amd64.tar.gz 1.5.0   # without internet
```

`update.sh` fetches the new version, takes over a newer `docker-compose.yml` (the old one is kept as `docker-compose.yml.alt-…`), starts the container and waits for the health check. If the new version does not start, it restores the previous version together with its database backup; `.env` then holds the previous version again. It runs with `/bin/sh` , also on a Synology. `--cleanup` never touches `data/` , backups, `.env` or the current `docker-compose.yml` .

- **`docker compose up` fails:** error message, exit code 1, `.env` and `docker-compose.yml` are restored. If there is a previous version, `update.sh` first asks whether to start it again (automatically with `-y`).
- **Still starting:** if the new version is still “starting” after `HELFI_UPDATE_WAIT` , `update.sh` asks whether to wait once more (automatically once with `-y`).
- **Database not restored:** `update.sh` says so explicitly, e.g. if the backups are not readable (then `sudo ./update.sh -y`) or there is no backup or only an incomplete one.
- **`--offline` without a version:** the version is taken from the image and recorded in `.env` , otherwise `update.sh` aborts. A relative path is taken from the current folder.
- `HELFI_UPDATE_WAIT` (time to wait, default 120 seconds) and `HELFI_IMAGE_REPO` are set on the command line, not in `.env` , e.g. `HELFI_UPDATE_WAIT=300 ./update.sh` . For your own registry, also change `image:` in `docker-compose.yml` ; it then counts as customized and is no longer replaced automatically.

> After an update the browser loads the new interface immediately (from version 1.5.0). If an older version still shows the old interface, press `Ctrl` + `Shift` + `R` once.

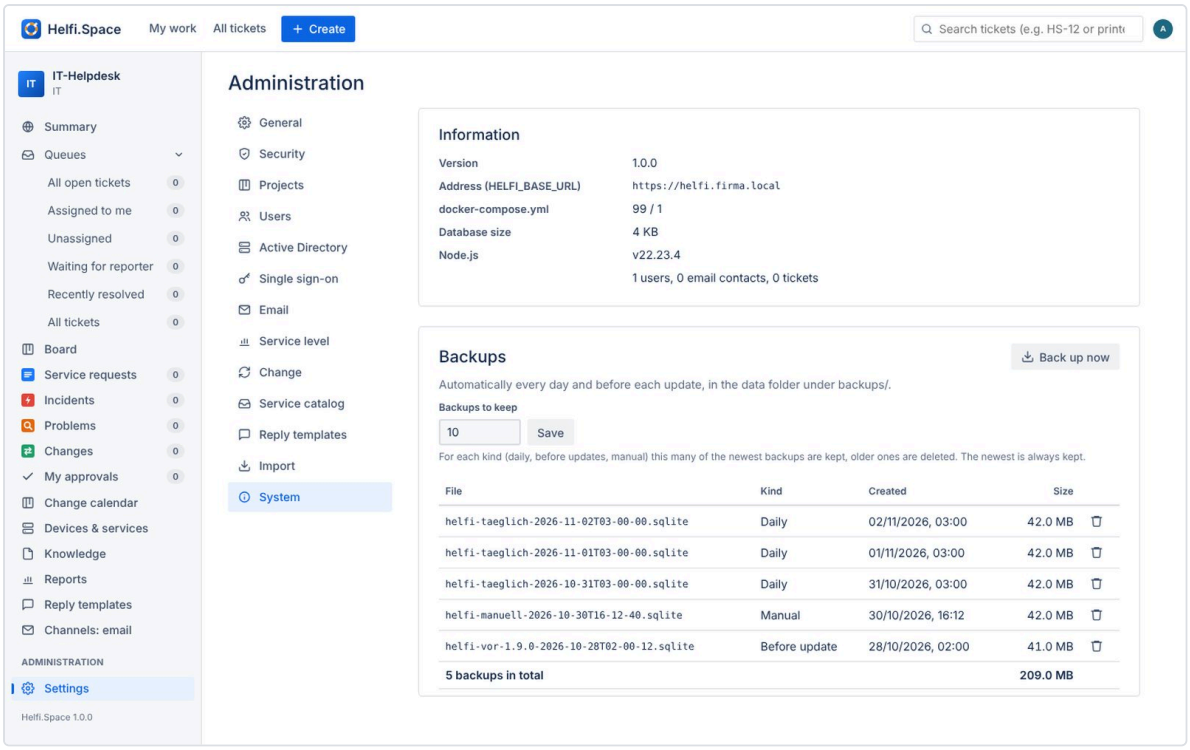
6.2 Pinning a version and going back

1. Enter a fixed version in `.env`, e.g. `HELFI_VERSION=1.4.0`.
2. If the newer version has already migrated the database, the older one does not start and names the matching backup in the log. Then:

```
docker compose down
cp data/helfi.sqlite data/helfi.sqlite.new
cp data/backups/helfi-vor-1.5.0-<date>.sqlite data/helfi.sqlite
rm -f data/helfi.sqlite-wal data/helfi.sqlite-shm
docker compose up -d
```

6.3 Backup and restore

- **Automatic:** daily, before every version change, before deleting a project and before every import starts, in `data/backups/`. A refused project deletion creates no backup. **Manual:** *Administration → System → Back up now*.
- **Backups to keep** (default 10): this many of the newest backups are kept per kind – daily, manual, before update, before project deletion and before import. Older ones are deleted when saving and from then on, at least the newest of each kind always stays. The list below shows kind, date, size and the total space used; 🗑️ deletes a single backup.
- Each backup is first written as a `.part` file and only renamed once complete. A `.part` file left behind after a crash is never used and is removed by the next clean-up after one hour.
- The backups contain only the database. **Always back up the whole `data/` folder** to another drive – including attachments, logo and `secret.key`.
- Without `secret.key` (or the `HELFI_SECRET` you set) stored passwords and 2FA keys can no longer be decrypted.



System: version, database, backups

6.4 Security

- Passwords of local accounts are stored with scrypt, stored credentials (service account, mailbox, SSO) are encrypted with AES-256-GCM and never sent to the browser.
- Anyone who changes server, port, encryption, service account or user, issuer or client ID for AD, IMAP, SMTP or SSO, weakens the certificate check (including “Fetched certificate” → “System certificate authorities”) or stores a different AD certificate, must enter the password or secret again in the same step, otherwise nothing is saved. This way a stored secret never goes to a different server; connection tests with a weakened check do not send it either.
- Sessions end after `HELFI_SESSION_HOURS` (default 12 hours) without use and at the latest 30 days after sign-in. Disabled accounts get the same message at sign-in as for a wrong password.
- Two-factor authentication per RFC 6238, each code works once, recovery codes are stored only as hashes.
- Brake on wrong passwords, cookies `HttpOnly` / `SameSite`, CSRF protection, strict Content Security Policy. The interface loads nothing from the internet.
- HTML emails are cleaned and shown in an isolated frame without scripts.
- The container runs without root rights, with a read-only program part, without additional Linux capabilities and with `no-new-privileges`.
- Tickets are never deleted; every change is recorded in the history.

6.5 Questions and solutions

Problem	Solution
Emails from Proxmox, NAS or monitoring are skipped	Add a sender rule “Create ticket, never reply”. The exact sender address is listed under *Email → Recently skipped emails*.
Every request counts as “from outside”	Set `HELFI_TRUST_PROXY` to the address of the reverse proxy; check the internal networks. The *Security* page shows how your address is classified.
Lost the phone with the authenticator	Sign in with a recovery code or use *Users → Reset 2FA*; then set it up again inside the company network.
AD sign-in stopped working, certificate notice	The DC certificate was renewed: *Active Directory → Fetch certificate* and accept it (or store the certificate authority).
An AD user’s role is wrong	Check the group membership in AD; if the group name is ambiguous, enter the full DN. With “Role locked” the role from Helfi.Space applies.
Notice “docker-compose.yml is outdated”	`./update.sh` takes over the new file, or download it from the release.
Helfi.Space does not start after a restore	Does `secret.key` or `HELFI_SECRET` match the database? The log (`docker compose logs helfi`) states the cause.
Setup code lost	It is in the log: `docker compose logs helfi` .